

Sri Lanka Institute of Information Technology



IT3021 – Data Warehousing & Business Intelligence

Assignment 01

IT23714434
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Step 1: Data set selection

The selected dataset is a Car Rental Transaction dataset representing the operations of a car rental company in Sri Lanka. Each record represents a single rental transaction and includes details related to customers, vehicles, branches, staff, insurance, and financial information.

The dataset covers an 18-month period, which exceeds the minimum requirement of one year. Additionally, the dataset was extended from 60,000 to 100,000 records using meaningful data generation techniques while maintaining proper relationships between entities.

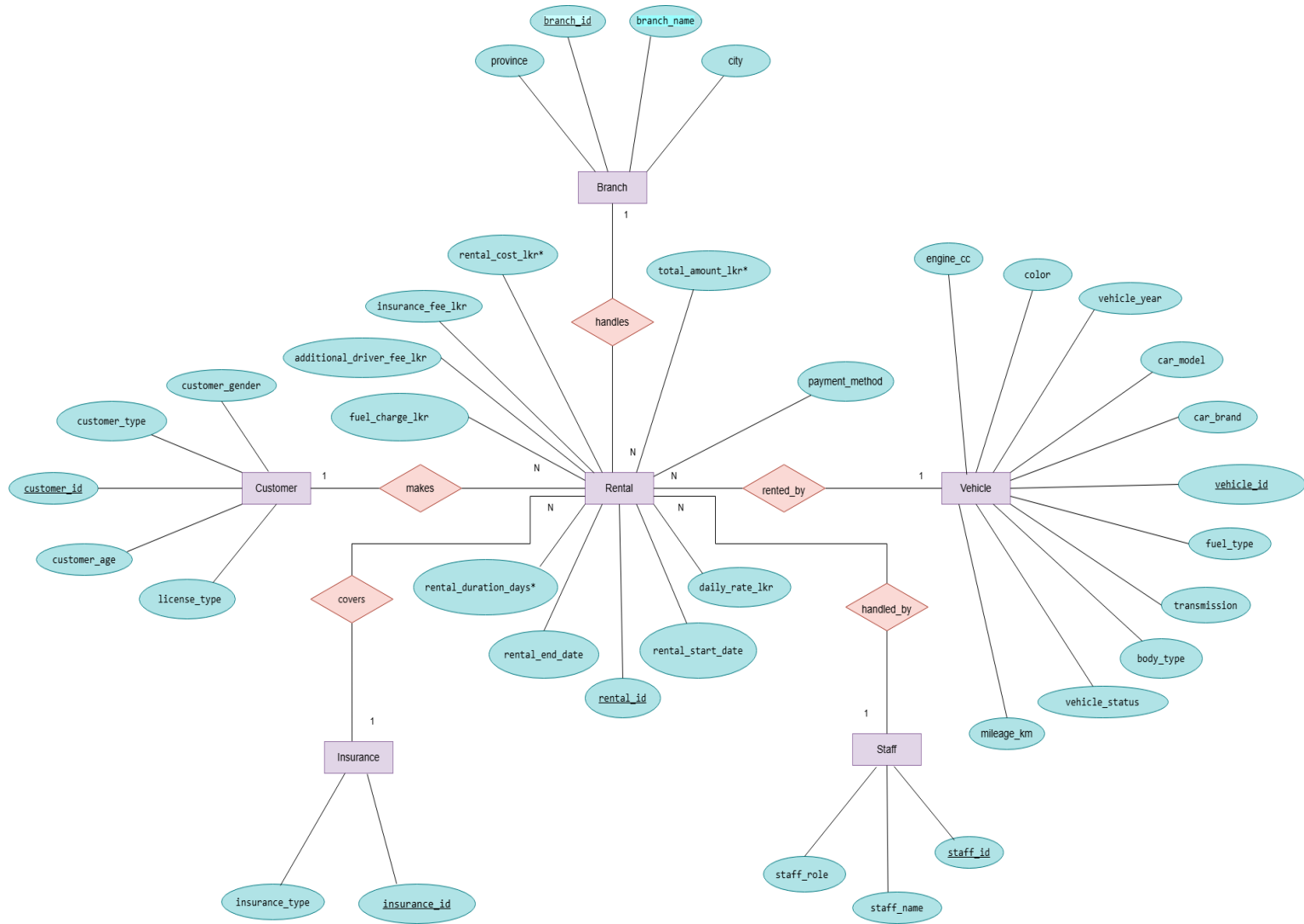
Data Set: Car_Rental

<https://www.kaggle.com/datasets/mrnize/car-rental>

The screenshot shows the Kaggle dataset page for 'Car_Rental'. The browser address bar displays 'kaggle.com/datasets/mrnize/car-rental'. The page title is 'Car_Rental' with a search bar and a 'Download' button. Below the title, there are tabs for 'Data Card', 'Code (0)', 'Discussion (0)', and 'Suggestions (0)'. The main content area shows the dataset 'Car_Rental_SLStyle_18Months.csv' (13.53 MB) with a 'Data Explorer' sidebar on the right. The 'Data Explorer' sidebar includes a 'Summary' section showing '1 file' and '30 columns'. The main table displays the first few rows of the dataset, with columns for 'Rental_ID', 'Rental_Start_Date', 'Rental_End_Date', '# Rental_Duration...', and 'Branch_Name'. The first row shows a rental starting on 2022-01-01 and ending on 2023-06-30, with a duration of 14 days, at the 'Negombo Airport...' branch. The second row shows a rental starting on 2022-11-02 and ending on 2022-11-13, with a duration of 11 days, at the 'Jaffna Auto Rent' branch. The third row shows a rental starting on 2022-09-30 and ending on 2022-10-05, with a duration of 5 days, at the 'Negombo Airport Rentals' branch.

Rental_ID	Rental_Start_Date	Rental_End_Date	# Rental_Duration...	Branch_Name
1	2022-01-01	2023-06-30	14	Negombo Airport...
1	2022-11-02	2022-11-13	11	Jaffna Auto Rent
2	2022-09-30	2022-10-05	5	Negombo Airport Rentals

ER Diagram :



* = Derived attribute (calculated from other fields)

Step 2: Preparation of data sources

The selected dataset was originally in a flat structure, containing customer, vehicle, rental, staff, branch and insurance related data in a single file.

The dataset was divided into multiple sources using a combination of:

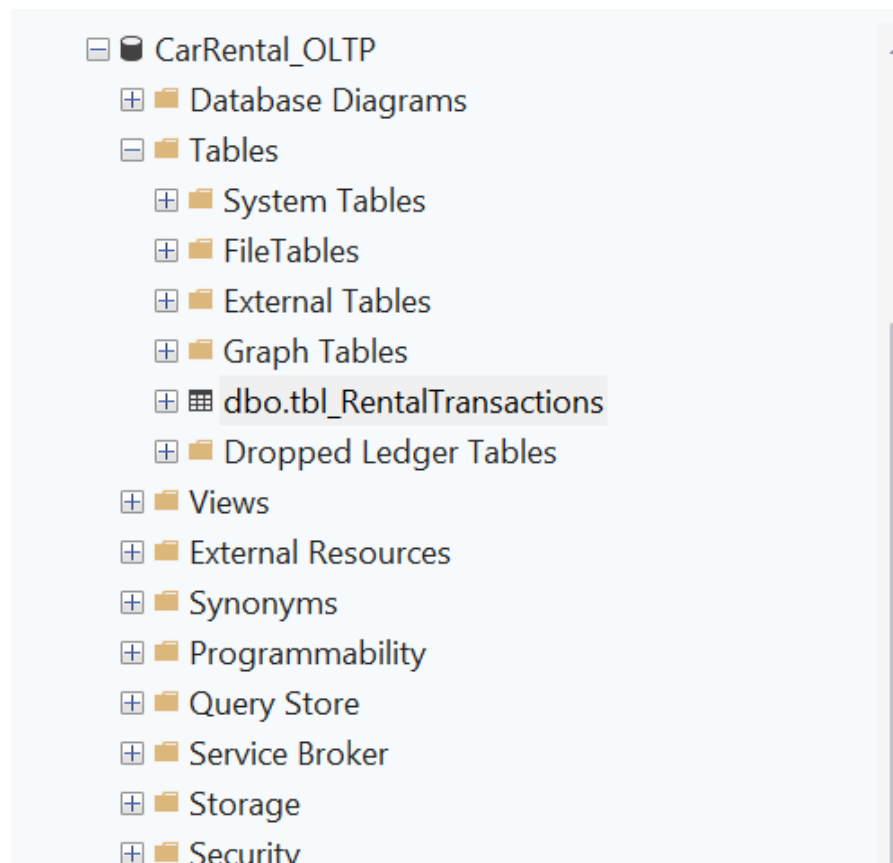
- * SQL database table
- * CSV files

Each source represents a specific entity identified in the ER diagram.

Main transactional table was created in the SQL database. Rental table stores the main transactional data related to vehicle rentals. Each record represents a single rental transaction. Customer data that contains unique customer records was stored as a CSV file. Insurance data that contains types of insurance (Basic, Standard, Premium) was stored as a CSV file. Vehicle data that contains vehicle details was stored as a CSV file. Staff data that contains staff identifiers was stored as a CSV file. Branch data that contains unique branch details was stored as a CSV file.

The dataset was successfully transformed from a flat structure into multiple organized data sources.

- * Transactional data was stored in SQL table
- * Master data was stored in CSV format
- * Relationships were maintained using primary and foreign keys
- * A hierarchy was included to support future data analysis



CarRental Data Source

The screenshot shows an Excel spreadsheet titled 'Customers - Excel'. The spreadsheet contains a list of customer data. The first row is a header with columns: Customer_ID, Customer, Customer, Customer, License_Type. The data rows show customer IDs (e.g., CUST-000, CUST-001, CUST-002), customer names (e.g., Corporate Female), and license types (e.g., International, Sri Lankan). The status bar at the bottom indicates 'Ready' and 'Scroll Lock'.

Customer_ID	Customer	Customer	Customer	License_Type
1	CUST-000	Corporate Female	18	International
2	CUST-000	Corporate Female	18	Sri Lankan
3	CUST-000	Corporate Female	19	Sri Lankan
4	CUST-000	Corporate Female	19	International
5	CUST-000	Corporate Female	20	International
6	CUST-000	Corporate Female	20	Sri Lankan
7	CUST-000	Corporate Female	21	Sri Lankan
8	CUST-000	Corporate Female	21	International
9	CUST-000	Corporate Female	22	Sri Lankan
10	CUST-001	Corporate Female	22	International
11	CUST-001	Corporate Female	23	Sri Lankan
12	CUST-001	Corporate Female	23	International
13	CUST-001	Corporate Female	24	International
14	CUST-001	Corporate Female	24	Sri Lankan
15	CUST-001	Corporate Female	25	Sri Lankan
16	CUST-001	Corporate Female	25	International
17	CUST-001	Corporate Female	26	Sri Lankan
18	CUST-001	Corporate Female	26	International
19	CUST-001	Corporate Female	27	International
20	CUST-002	Corporate Female	27	Sri Lankan

Customers.csv

Insurance - Excel																		
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A1 Insurance_ID																		
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1	Insurance	Insurance_Type																
2	INS-001	Basic																
3	INS-002	Full Coverage																
4	INS-003	Premium																
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Insurance.csv

Vehicle.csv - Excel																		
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A1 Vehicle_ID																		
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1	Vehicle_ID	Car_Branc	Car_Mode	Vehicle_Yr	Engine_CC	Fuel_Type	Transmiss	Mileage_K	Body_Typ	Color	Vehicle_Status							
2	VEH-0001	Honda	CR-V	2012	1800	Diesel	Manual	118358	Sedan	Black	Available							
3	VEH-0002	Honda	CR-V	2012	2500	Diesel	Automatic	120468	Pickup	Blue	Available							
4	VEH-0003	Honda	CR-V	2012	2500	Petrol	Manual	125542	Hatchback	Red	Maintenance							
5	VEH-0004	Honda	CR-V	2012	2500	Diesel	Automatic	112204	Hatchback	Silver	Maintenance							
6	VEH-0005	Honda	CR-V	2012	1800	Petrol	Manual	122975	Sedan	White	Rented							
7	VEH-0006	Honda	CR-V	2013	1300	Hybrid	Automatic	114296	SUV	Black	Rented							
8	VEH-0007	Honda	CR-V	2013	1000	Petrol	Manual	121732	Hatchback	Blue	Rented							
9	VEH-0008	Honda	CR-V	2013	2500	Petrol	Automatic	111372	Pickup	Red	Rented							
10	VEH-0009	Honda	CR-V	2013	1500	Petrol	Manual	110901	Sedan	Silver	Maintenance							
11	VEH-0010	Honda	CR-V	2013	2500	Hybrid	Manual	108044	Sedan	White	Available							
12	VEH-0011	Honda	CR-V	2014	1000	Petrol	Manual	110382	Hatchback	Black	Rented							
13	VEH-0012	Honda	CR-V	2014	2500	Petrol	Automatic	109047	SUV	Blue	Available							
14	VEH-0013	Honda	CR-V	2014	1300	Hybrid	Automatic	106161	Sedan	Red	Maintenance							
15	VEH-0014	Honda	CR-V	2014	2500	Hybrid	Manual	111350	Pickup	Silver	Maintenance							
16	VEH-0015	Honda	CR-V	2014	2000	Petrol	Automatic	121804	Hatchback	White	Rented							
17	VEH-0016	Honda	CR-V	2015	1800	Petrol	Manual	101517	Pickup	Black	Available							
18	VEH-0017	Honda	CR-V	2015	1300	Petrol	Manual	101095	Pickup	Blue	Available							
19	VEH-0018	Honda	CR-V	2015	1300	Diesel	Automatic	101410	SUV	Red	Rented							
20	VEH-0019	Honda	CR-V	2015	2000	Diesel	Manual	96447	Sedan	Silver	Available							
21	VEH-0020	Honda	CR-V	2015	1800	Hybrid	Automatic	113575	Hatchback	White	Rented							

Vehicle.csv

Branch.csv - Excel

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Branch_ID

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
1	Branch_ID	Branch_N	City	Province															
2	BR-001	Colombo	(Colombo)	Western															
3	BR-002	Kandy	Drii Kandy	Central															
4	BR-003	Galle	Coas Galle	Southern															
5	BR-004	Negombo	Negombo	Western															
6	BR-005	Jaffna	Aut Jaffna	Northern															
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Branch

Branch.csv

Staff.csv - Excel

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POSSIBLE DATA LOSS Some features might be lost if you save this workbook in the comma-delimited (.csv) format. To preserve these features, save it in an Excel file format. Don't show again Save As...

Staff_ID

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
1	Staff_ID	Staff_Nam	Staff_Role	Branch_ID															
2	1	Amara Wi	Senior Rer	BR-001															
3	2	Kasun Ban	Rental Age	BR-001															
4	3	Nimesha J	Branch M	BR-002															
5	4	Tharindu I	Rental Age	BR-002															
6	5	Sachini Gu	Senior Rer	BR-003															
7	6	Ravindu N	Rental Age	BR-003															
8	7	Dilani Pat	Branch M	BR-001															
9	8	Harsha Ra	Fleet Coor	BR-004															
10	9	Piyumi Se	Rental Age	BR-004															
11	10	Janith Pe	Branch M	BR-004															
12	11	Ayesha Fe	Senior Rer	BR-005															
13	12	Sanjaya K	Rental Age	BR-005															
14	13	Madushi F	Fleet Coor	BR-002															
15	14	Roshan Ja	Rental Age	BR-003															
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Staff

Staff.csv

- ❑ **Customer** – Details about the customers including personal information
- ❑ **Vehicle** – Details about vehicles such as type, model, availability, and status
- ❑ **Rental** – Records of rental transactions including rental dates, duration, and cost
- ❑ **Branch** – Details of rental branches including location and branch information
- ❑ **Staff** – Information about employees who handle rental operations
- ❑ **Insurance** – Details of insurance policies associated with vehicles

Step 3: Solution architecture

Using different processes, architectures, and technologies we can manage data from various sources and convert them to business insights to make decisions, analysis data and report building.

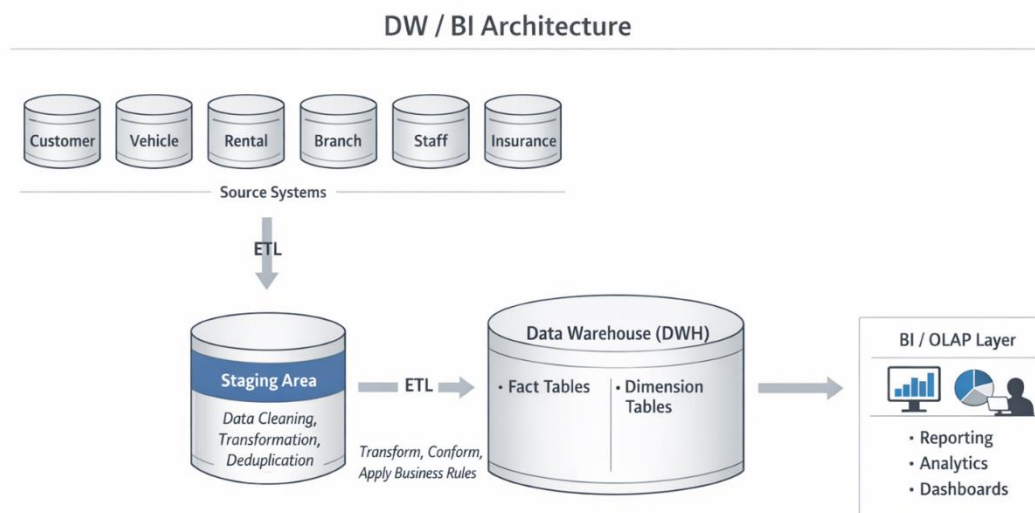
In this solution, there are 3 main components up to the ETL process in the process of Datawarehouse design.

- Source Files
- Staging DB
- Datawarehouse

Source Files : There are 7 source tables in three different format as the data sources.

Staging DB : Extracted all source details and change the data types of the attributes that are in source file to relevant data type and loaded that details into the staging database.

Datawarehouse : After the staging, the information in staging database will become the sources to the transformation process. In this process data will be transformed & loaded into the tables in Datawarehouse database.



Step 4: Data warehouse design & development

The data warehouse follows a **star schema**, a single fact table (FactRental) surrounded by six dimension tables.

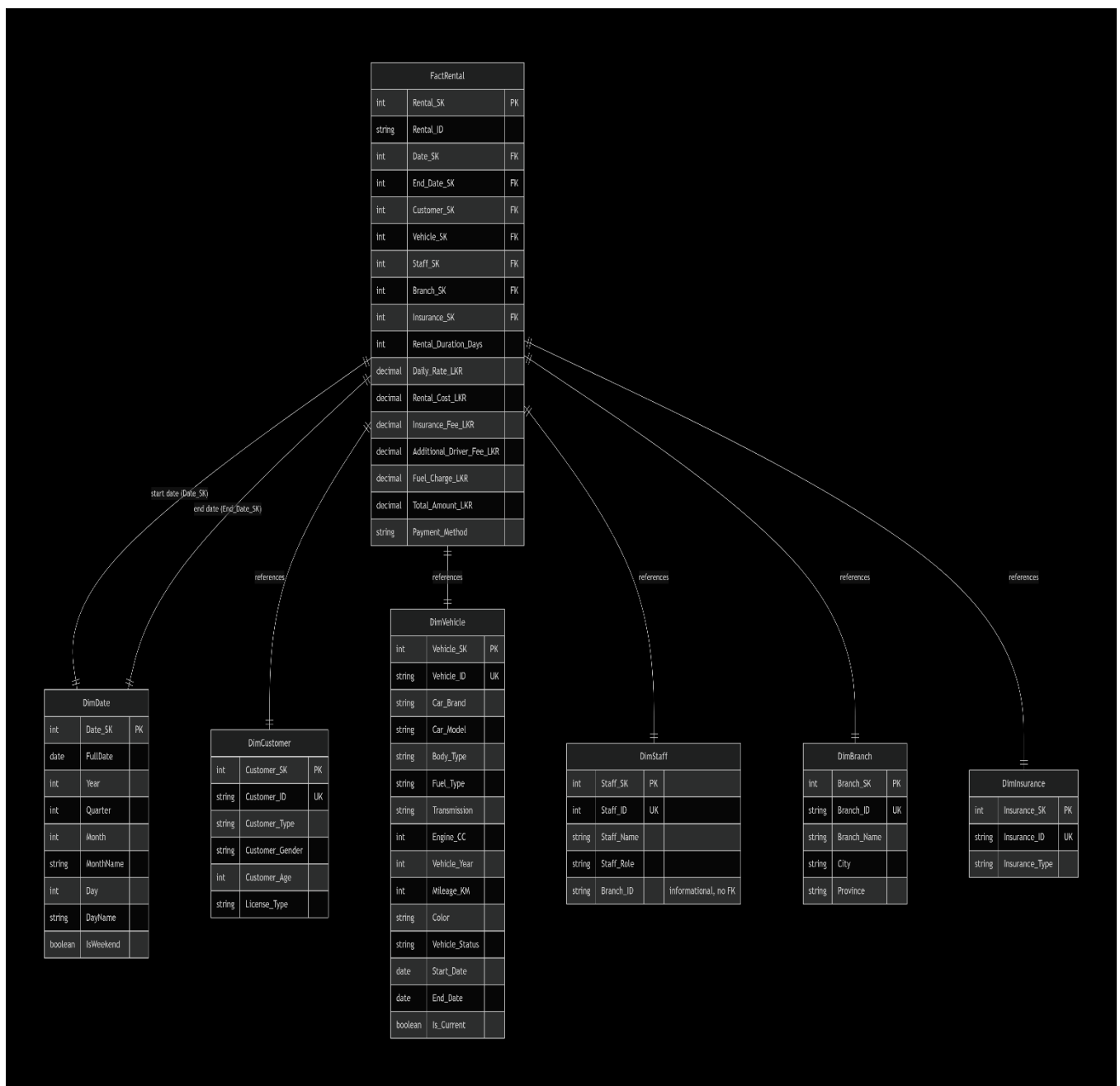
Fact Table: FactRental - stores each rental transaction at the most granular level (one row per rental).

Dimension Tables:

- **DimDate** - date attributes for start date and end date (two separate role playing dimensions)
- **DimCustomer** - customer demographic information
- **Dimvehicle** - vehicle details with **Type 2 Slowly Changing Dimension** (tracking historical changes)
- **DimBranch** - branch location details
- **DimInsurance** - insurance plan details
- **DimStaff** - staff member information

Assumptions

1. **Granularity of the fact table:** One row per rental transaction. No aggregation at the fact level.
2. **Slowly Changing Dimension:** DimVehicle is Type 2 because a vehicle's mileage and daily rate can change over time, and historical comparisons require tracking the version of the vehicle at the time of rental.



```

SQLQuery1.1...User (54)
1 BEGIN TRY
2 DROP TABLE [dbo].[DimDate]
3 END TRY
4
5 BEGIN CATCH
6 /*No Action*/
7 END CATCH
8
9 /*****
10
11 CREATE TABLE [dbo].[DimDate]
12 (
13 [DateKey] INT primary key,
14 [Date] DATETIME,
15 [FullDateUK] CHAR(10), -- Date in dd-MM-yyyy format
16 [FullDateUSA] CHAR(10), -- Date in MM-dd-yyyy format
17 [DayOfMonth] VARCHAR(2), -- Field will hold day number of Month
18 [DaySuffix] VARCHAR(4), -- Apply suffix as 1st, 2nd ,3rd etc
19 [DayName] VARCHAR(9), -- Contains name of the day, Sunday, Monday
20 [DayOfWeekUSA] CHAR(1), -- First Day Sunday=1 and Saturday=7
21 [DayOfWeekUK] CHAR(1), -- First Day Monday=1 and Sunday=7
22 [DayOfWeekInMonth] VARCHAR(2), --1st Monday or 2nd Monday in Month
23 [DayOfWeekInYear] VARCHAR(2),
24 [DayOfQuarter] VARCHAR(3),
25 [DayOfYear] VARCHAR(3),
26 [WeekOfMonth] VARCHAR(1), -- Week Number of Month
27 [WeekOfQuarter] VARCHAR(2), --Week Number of the Quarter
28 [WeekOfYear] VARCHAR(2), --Week Number of the Year
29 [Month] VARCHAR(2), --Number of the Month 1 to 12
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DESKTOP-US...DimCustomer  SQLQuery1.1...User (54))*
272
273 -- DimCustomer
274 CREATE TABLE DimCustomer (
275     Customer_SK INT IDENTITY(1,1) PRIMARY KEY,
276     Customer_ID VARCHAR(20) NOT NULL UNIQUE,
277     Customer_Type VARCHAR(50),
278     Customer_Gender CHAR(1),
279     Customer_Age INT,
280     License_Type VARCHAR(50)
281 );
282

```

Column Name	Data Type	Allow Nulls
Customer_SK	int	☐
Customer_ID	varchar(20)	☐
Customer_Type	varchar(50)	☒
Customer_Gender	char(1)	☒
Customer_Age	int	☒
License_Type	varchar(50)	☒
		☐

Column Properties

DimCustomer

```

244
245 -- DimVehicle (SCD Type 2)
246 CREATE TABLE DimVehicle (
247     Vehicle_SK INT IDENTITY(1,1) PRIMARY KEY,
248     Vehicle_ID VARCHAR(10) NOT NULL UNIQUE,
249     Car_Brand VARCHAR(50),
250     Car_Model VARCHAR(50),
251     Body_Type VARCHAR(50),
252     Fuel_Type VARCHAR(50),
253     Transmission VARCHAR(50),
254     Engine_CC INT,
255     Vehicle_Year INT,
256     Mileage_KM INT,
257     Color VARCHAR(30),
258     Vehicle_Status VARCHAR(20),
259     Start_Date DATE NOT NULL,
260     End_Date DATE NULL,
261     Is_Current BIT NOT NULL
262 );
263

```

Column Name	Data Type	Allow Nulls
Rental_SK	int	☐
Rental_ID	varchar(20)	☐
Date_SK	int	☐
End_Date_SK	int	☐
Customer_SK	int	☐
Vehicle_SK	int	☐
Staff_SK	int	☐
Branch_SK	int	☐
Insurance_SK	int	☐
Rental_Duration_Days	int	☒
Daily_Rate_LKR	decimal(10, 2)	☒
Rental_Cost_LKR	decimal(10, 2)	☒
Insurance_Fee_LKR	decimal(10, 2)	☒
Additional_Driver_Fee_LKR	decimal(10, 2)	☒
Fuel_Charge_LKR	decimal(10, 2)	☒
Total_Amount_LKR	decimal(10, 2)	☒
Payment_Method	varchar(50)	☒

Column Properties

DimVehicle


```
SQLQuery1.1...\User (54))*
263
264 -- DimBranch
265 CREATE TABLE DimBranch (
266     Branch_SK INT IDENTITY(1,1) PRIMARY KEY,
267     Branch_ID VARCHAR(10) NOT NULL UNIQUE,
268     Branch_Name VARCHAR(100),
269     City VARCHAR(50),
270     Province VARCHAR(50)
271 );
272
```

Column Name	Data Type	Allow Nulls
Branch_SK	int	☐
Branch_ID	varchar(10)	☐
Branch_Name	varchar(100)	☒
City	varchar(50)	☒
Province	varchar(50)	☒
		☐

DimBranch

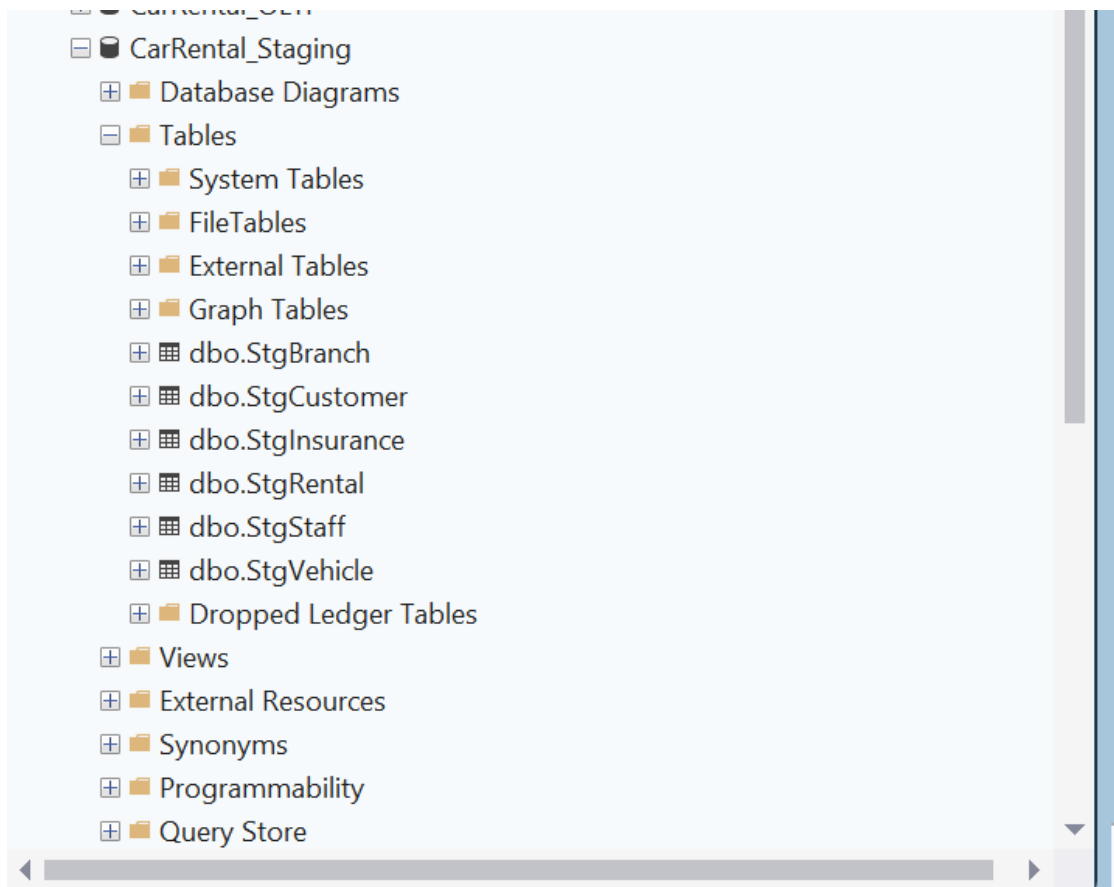
```
DESKTOP-USP...imInsurance
SQLQuery1.1...\User (54))*
291
292 -- DimInsurance
293 CREATE TABLE DimInsurance (
294     Insurance_SK INT IDENTITY(1,1) PRIMARY KEY,
295     Insurance_ID VARCHAR(10) NOT NULL UNIQUE,
296     Insurance_Type VARCHAR(50)
297 );
298
299
```

Column Name	Data Type	Allow Nulls
Insurance_SK	int	☐
Insurance_ID	varchar(10)	☐
Insurance_Type	varchar(50)	☒
		☐

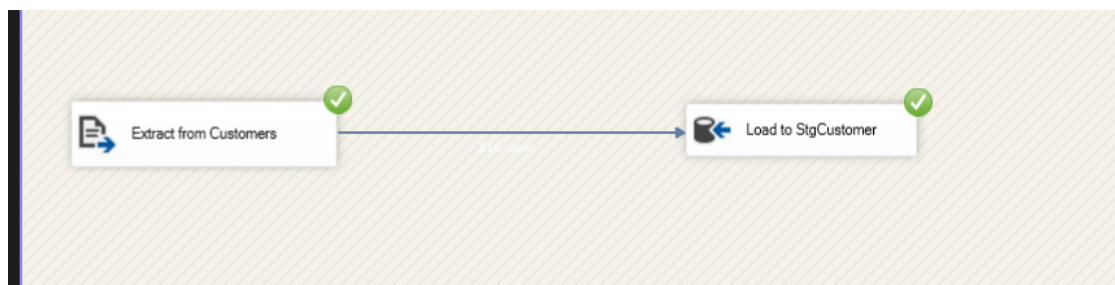
Column Properties

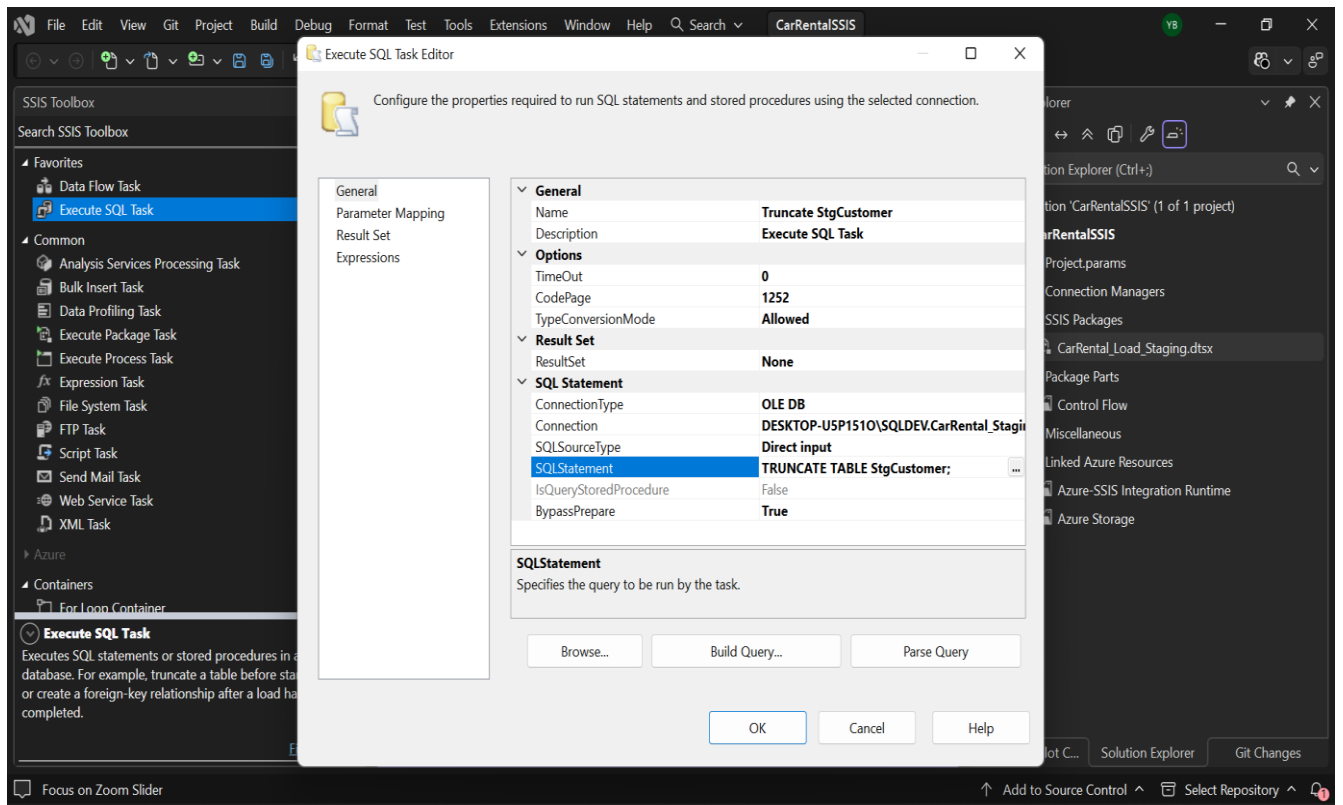
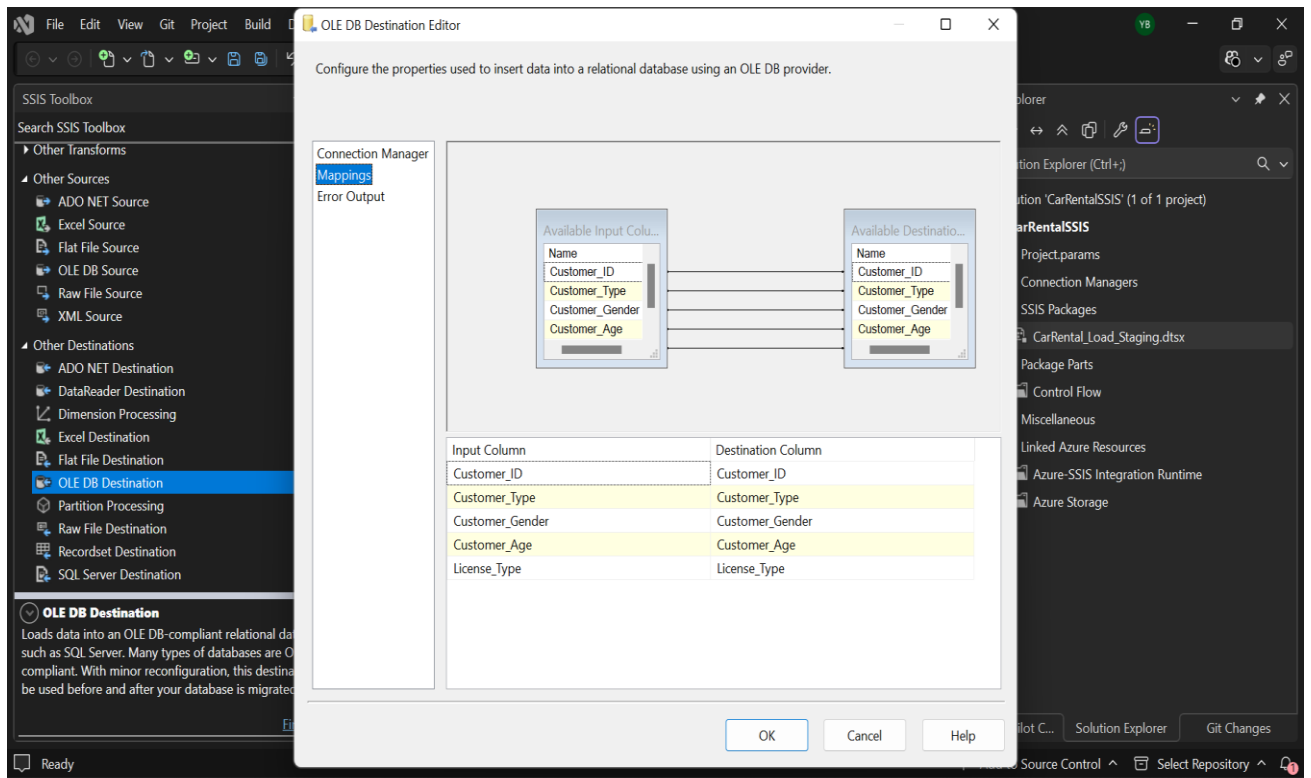
DimInsurance

First extracted all the data from tables which are in the CarRental_OLTP and flat files into separated staging DB called CarRental_Staging as shown in the below using SQL Server Integration Service Software.

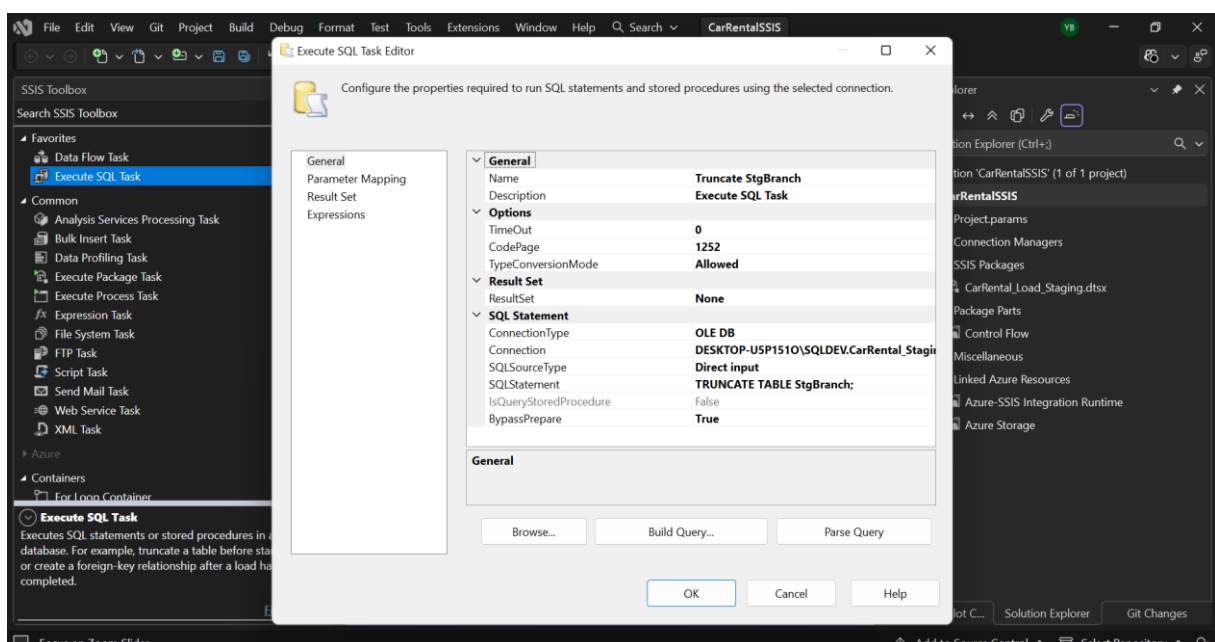
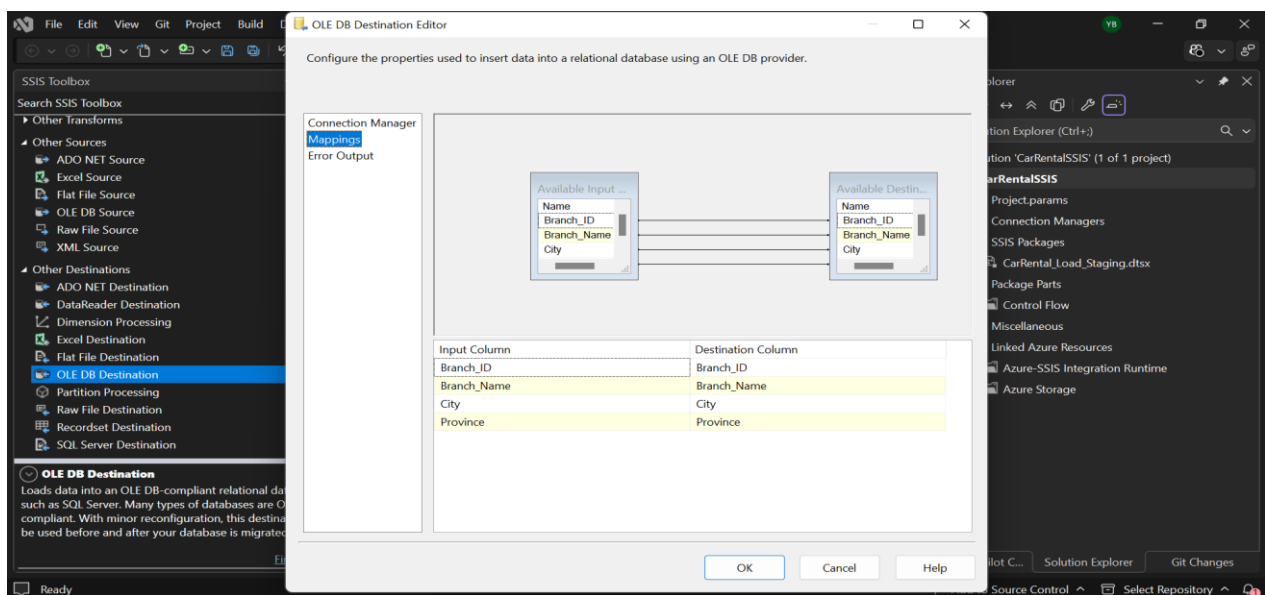
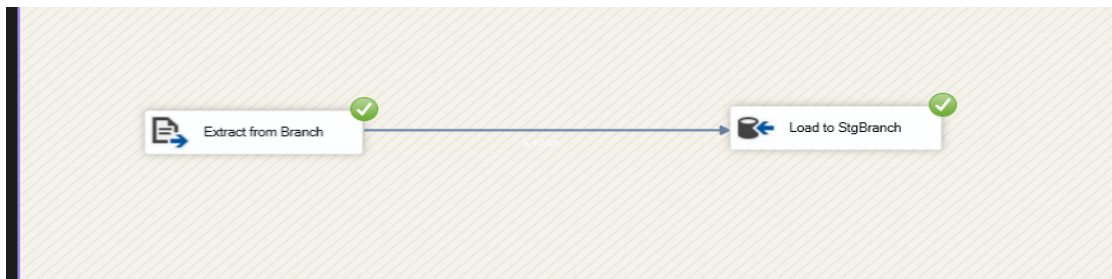


Extract Customer Details from Customers.csv and load into staging Customers table. Used Execute SQL Task SSIS tool Truncate table for SQL command as truncate table dbo.Customers in CarRental_Staging Database

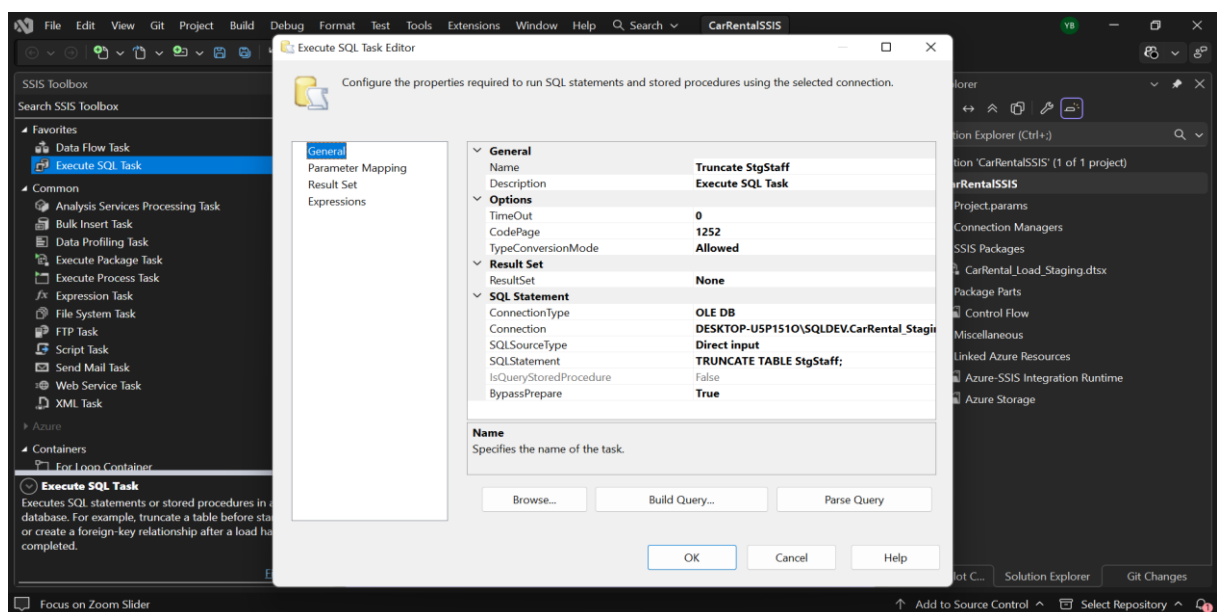
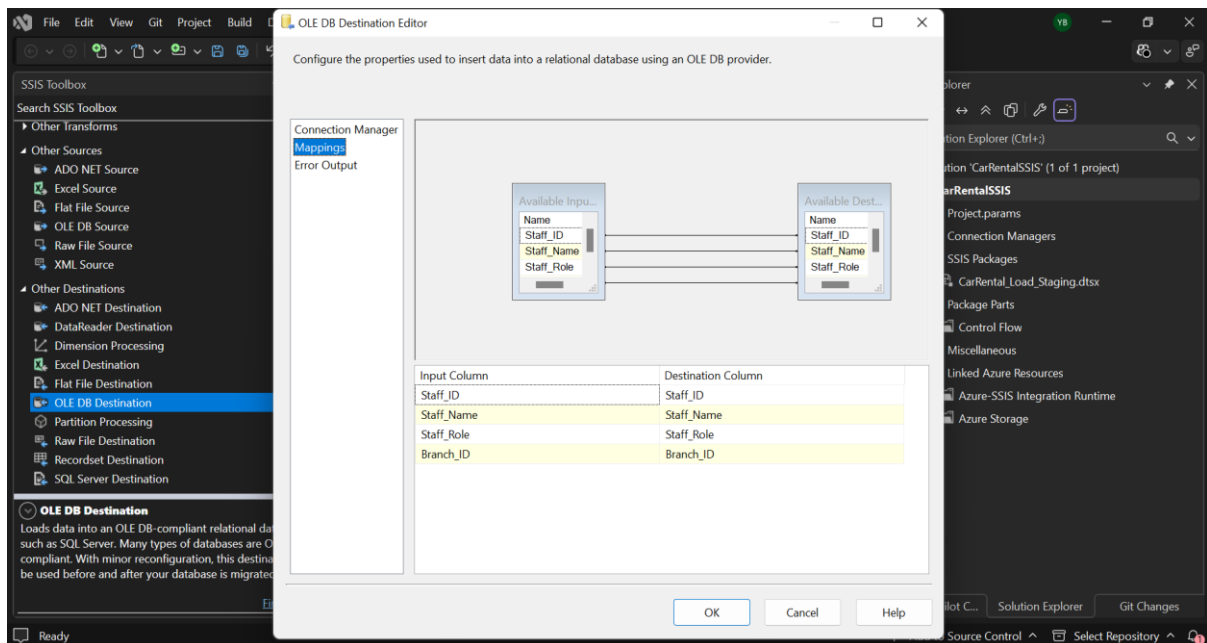
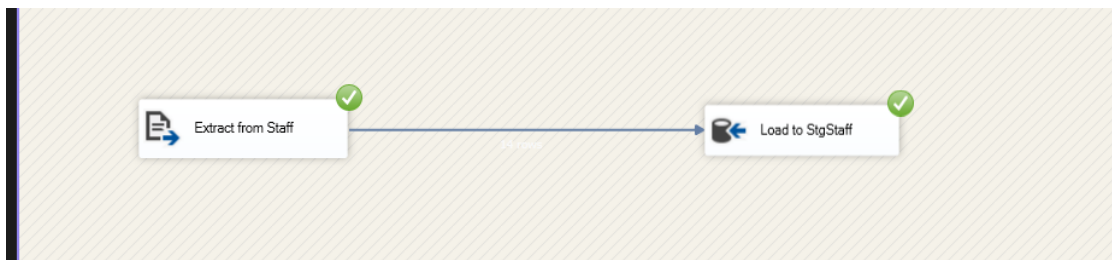




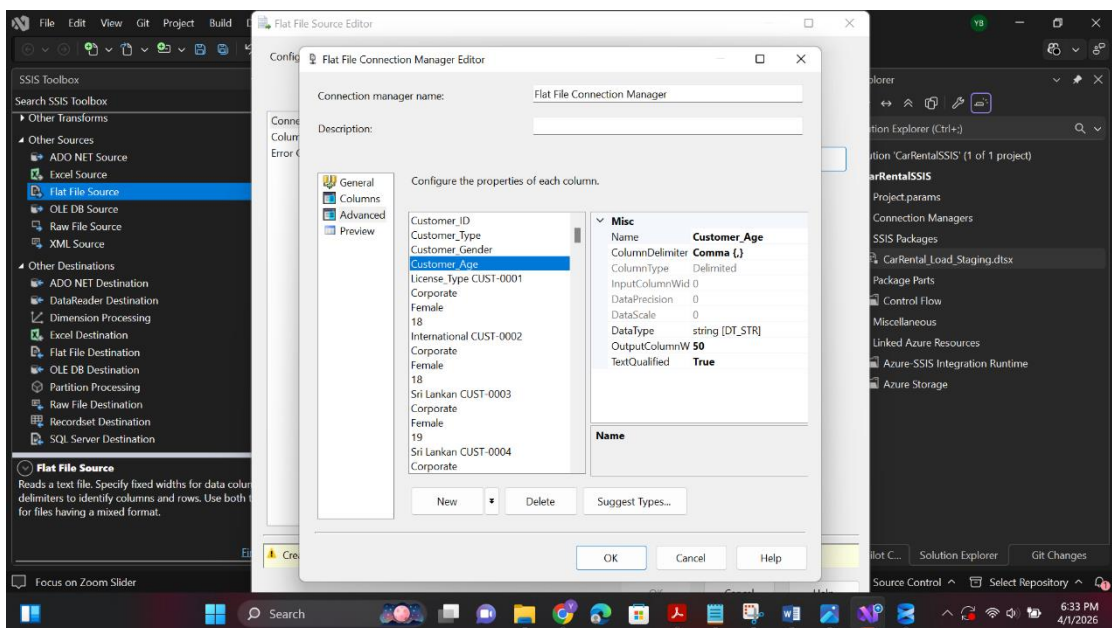
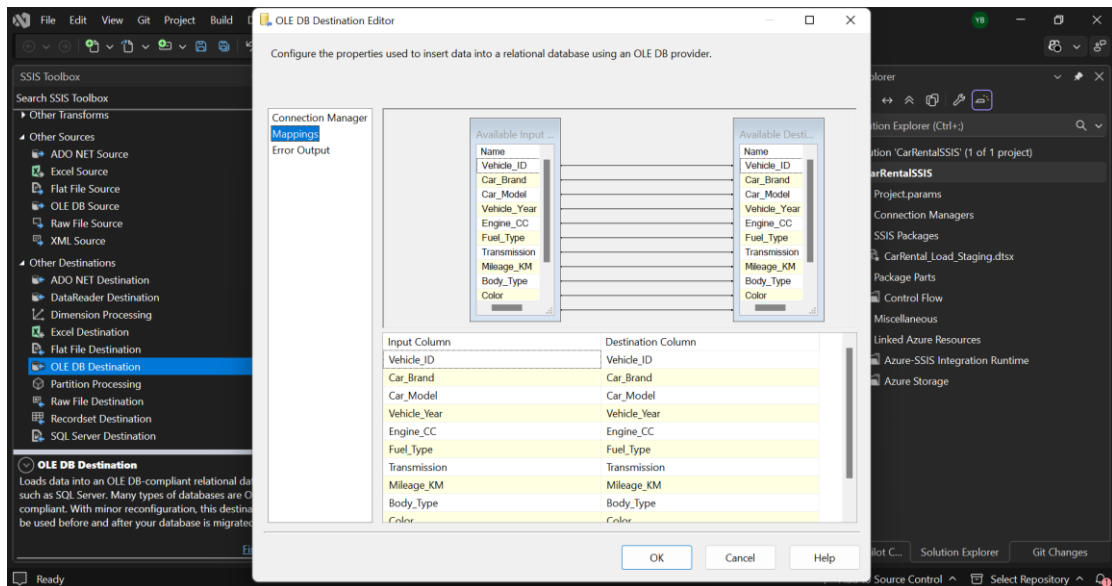
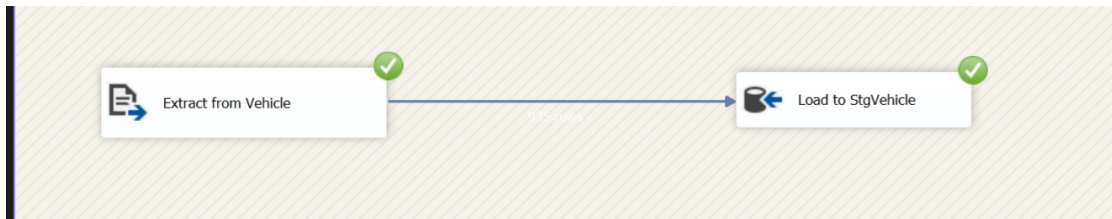
Extract Branch Details from Branch.csv and load into staging Branch table. Used Execute SQL Task SSIS tool Truncate table for SQL command as truncate table dbo.Branch in CarRental_Staging Database



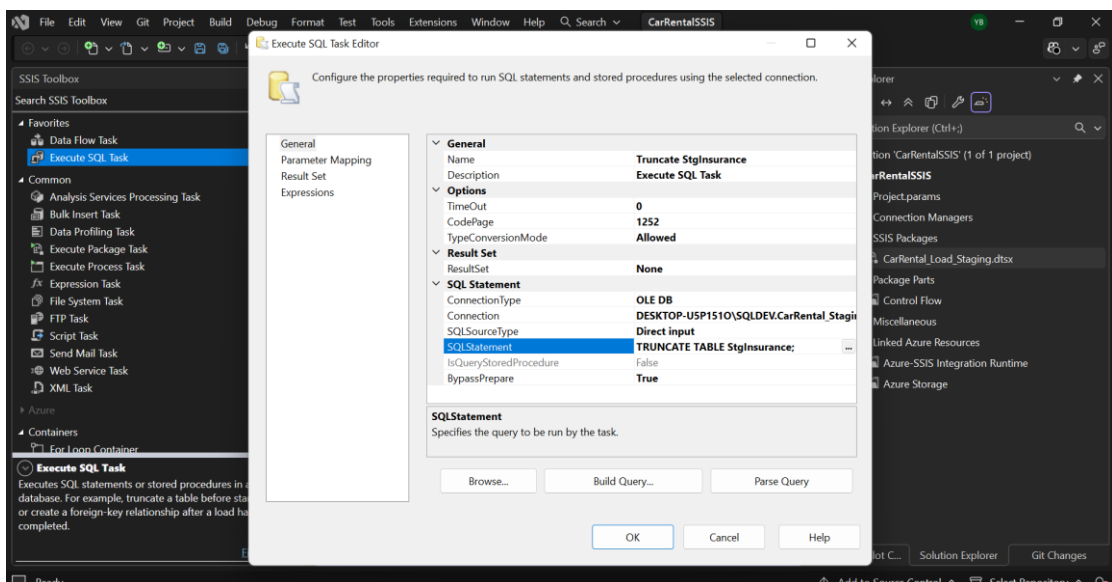
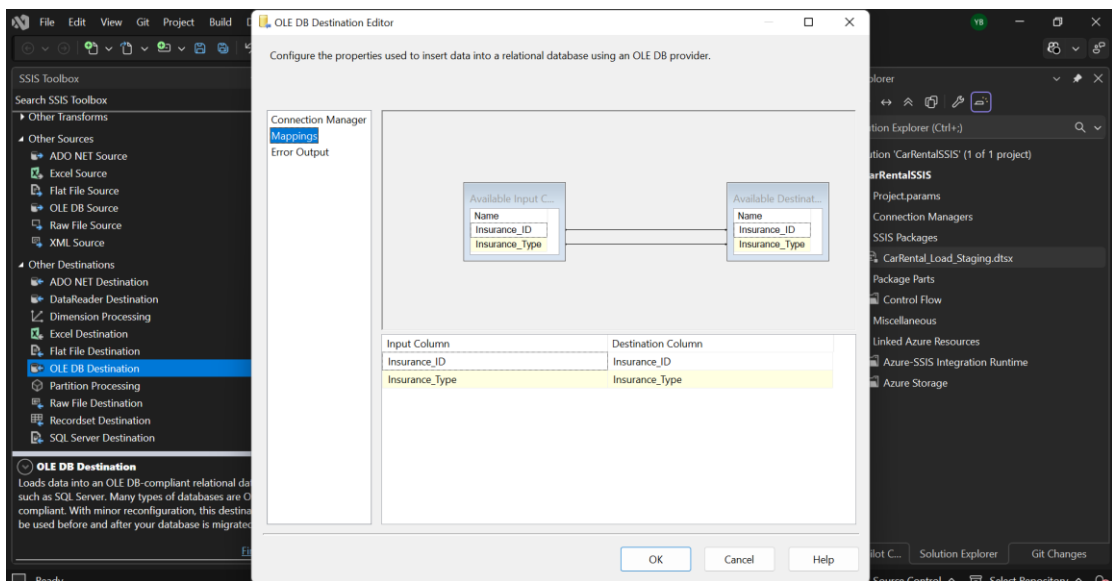
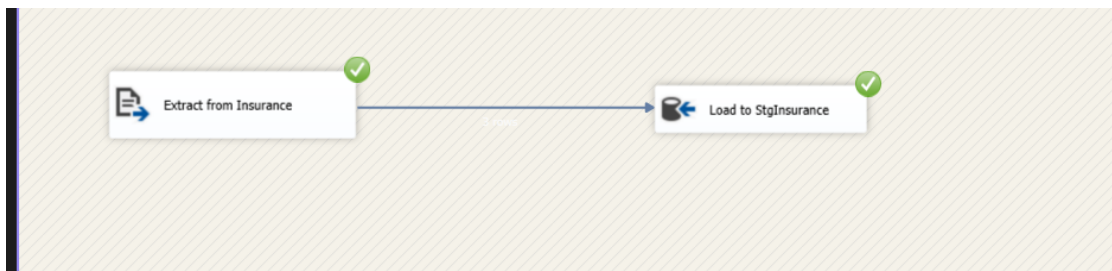
Extract Staff Details from Staff.csv and load into staging Staff table. Used Execute SQL Task SSIS tool Truncate table for SQL command as truncate table dbo.Staff in CarRental_Staging Database



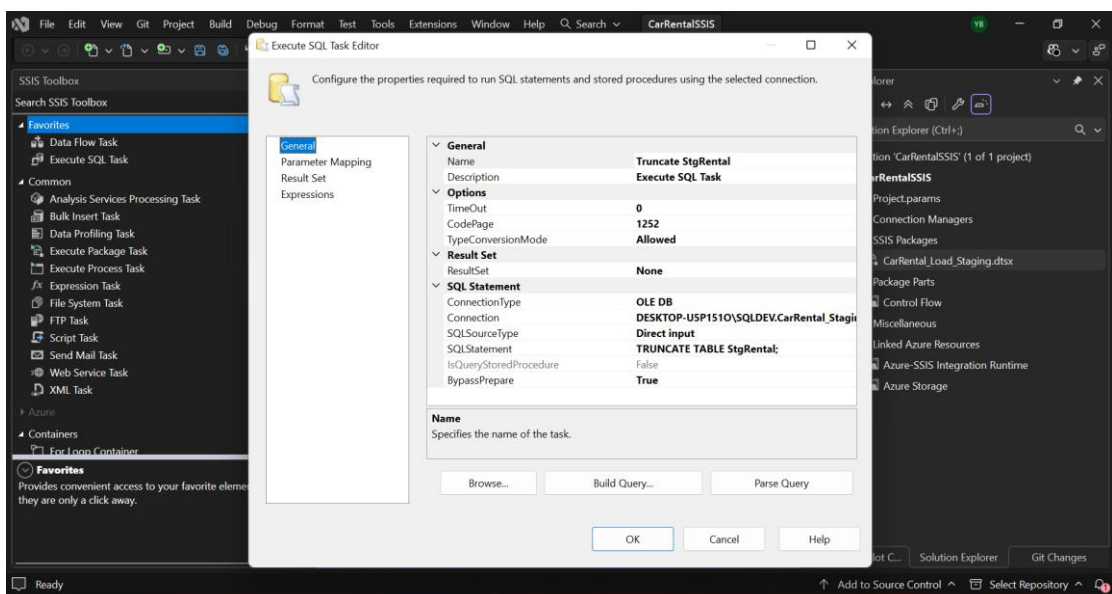
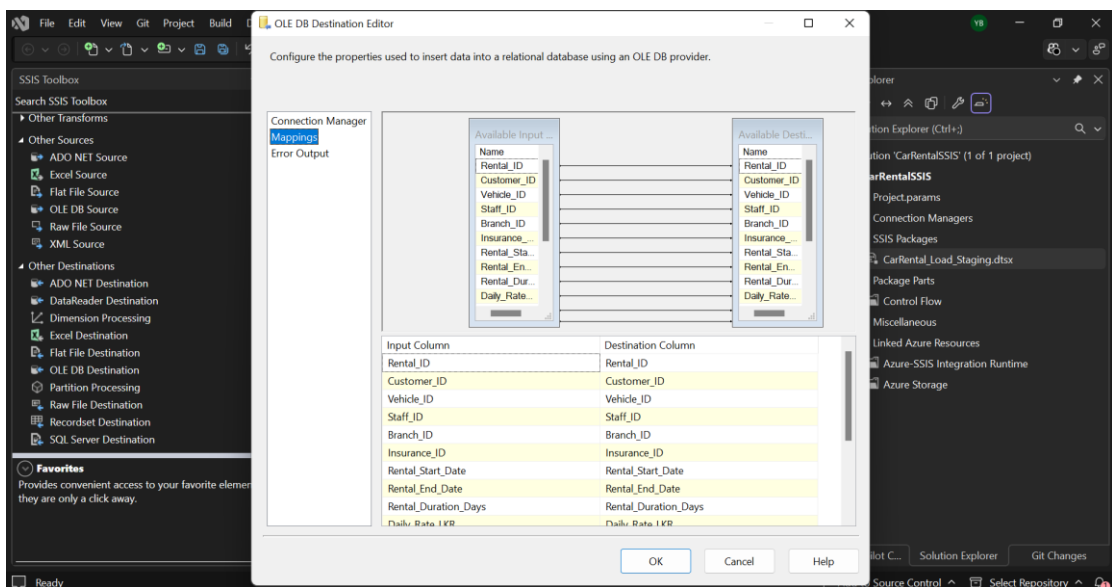
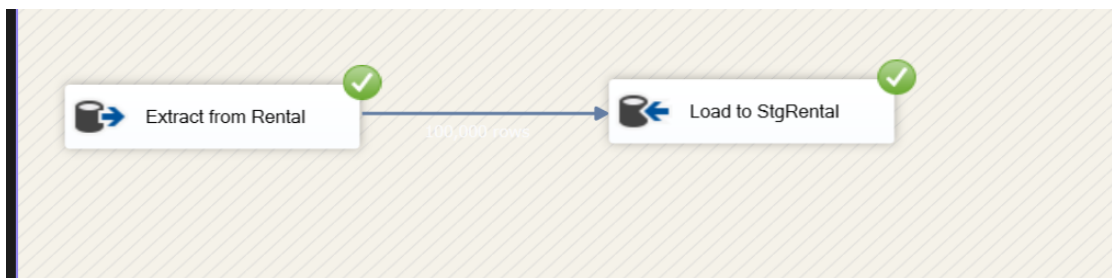
Extract Vehicle Details from Vehicle.csv and load into staging Vehicle table. Used Execute SQL Task SSIS tool Truncate table for SQL command as truncate table dbo.Vehicle in CarRental_Staging Database



Extract Insurance Details from Insurance.csv and load into staging Insurance table.
Used Execute SQL Task SSIS tool Truncate table for SQL command as truncate table
dbo.Insurance in CarRental_Staging Database



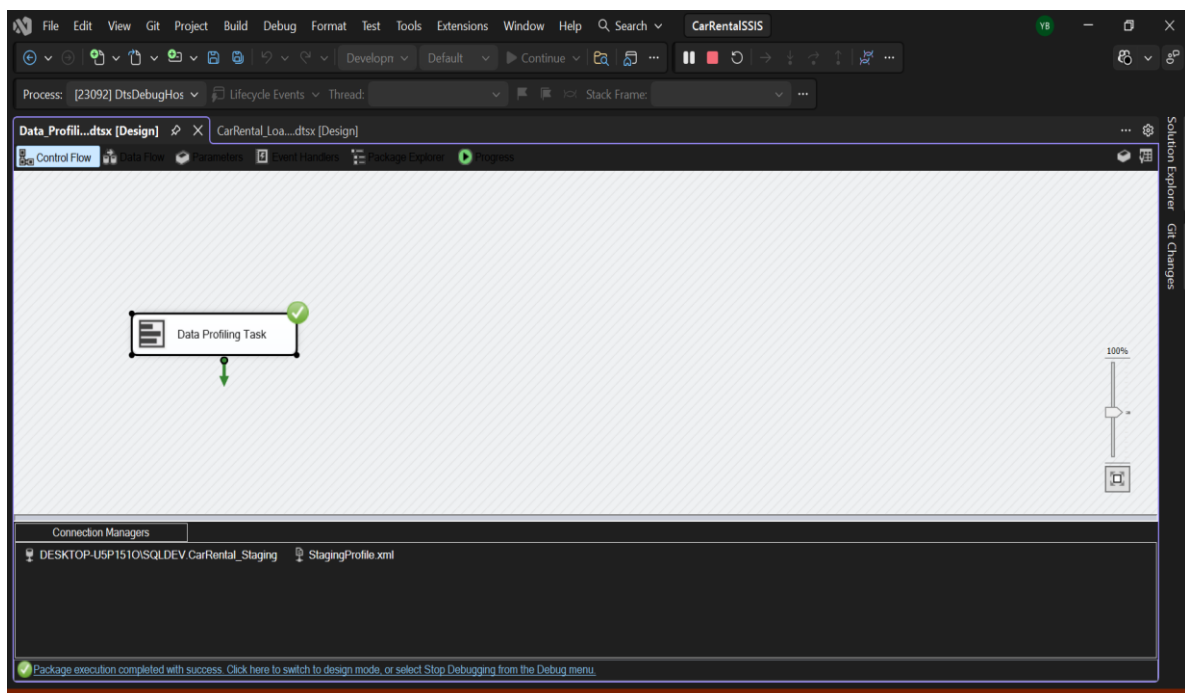
Extract Rental Details from Rental.csv and load into staging Rental table. Used Execute SQL Task SSIS tool Truncate table for SQL command as truncate table dbo.Rental in CarRental_Staging Database

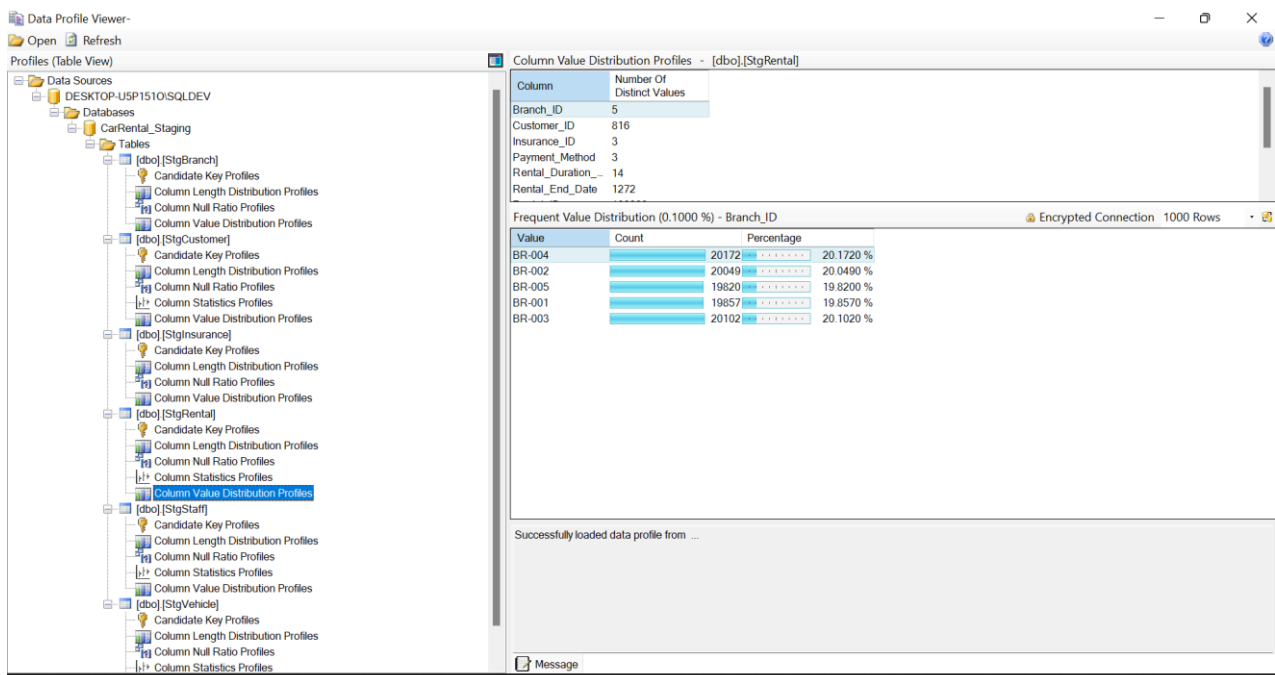
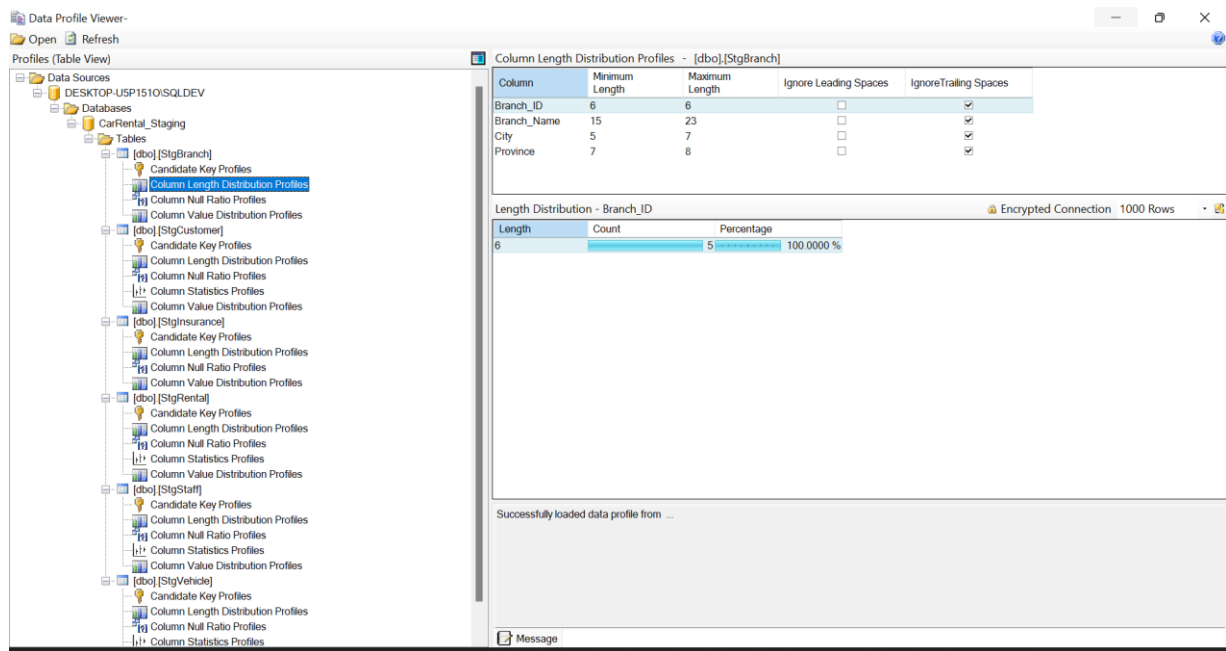


Data Profiling

Staging table data was used to analyzed and determined what types of transformation was needed to perform on the data.

- Right click on SSIS Packages and New SSIS Package selected.
- In the Control Flow of Data_Profiling.dtsx, drag and drop the Data Profiling Task and double click to open the configuration.
- Clicked on Quick Profile button to open Single Table Quick Profile Form.
- Clicked on New button and create the connection to CarRental_Staging
- Once the green tick appeared, double click the Data Profiling Task and Click on the Open Profile Viewer to view the analyzed data





Data Profile Viewer

Open Refresh

Profiles (Table View)

Data Sources

- DESKTOP-USP1510\SQLDEV
 - Databases
 - CarRental_Staging
 - Tables
 - [dbo].[StgBranch]
 - Candidate Key Profiles
 - Column Length Distribution Profiles
 - Column Null Ratio Profiles
 - Column Value Distribution Profiles
 - [dbo].[StgCustomer]
 - Candidate Key Profiles
 - Column Length Distribution Profiles
 - Column Null Ratio Profiles
 - Column Statistics Profiles
 - Column Value Distribution Profiles
 - [dbo].[StgInsurance]
 - Candidate Key Profiles
 - Column Length Distribution Profiles
 - Column Null Ratio Profiles
 - Column Value Distribution Profiles
 - [dbo].[StgRental]
 - Candidate Key Profiles
 - Column Length Distribution Profiles
 - Column Null Ratio Profiles
 - Column Statistics Profiles
 - Column Value Distribution Profiles
 - [dbo].[StgStaff]
 - Candidate Key Profiles
 - Column Length Distribution Profiles
 - Column Null Ratio Profiles
 - Column Statistics Profiles
 - Column Value Distribution Profiles
 - [dbo].[StgVehicle]
 - Candidate Key Profiles
 - Column Length Distribution Profiles
 - Column Null Ratio Profiles
 - Column Statistics Profiles

Column Null Ratio Profiles - [dbo].[StgInsurance]

Encrypted Connection 1000 Rows

Column	Null Count	Null Percentage
Insurance_ID	0	0.0000 %
Insurance_Type	0	0.0000 %

Successfully loaded data profile from ...

Message

Data Profile Viewer

Open Refresh

Profiles (Table View)

Data Sources

- DESKTOP-USP1510\SQLDEV
 - Databases
 - CarRental_Staging
 - Tables
 - [dbo].[StgBranch]
 - Candidate Key Profiles
 - Column Length Distribution Profiles
 - Column Null Ratio Profiles
 - Column Value Distribution Profiles
 - [dbo].[StgCustomer]
 - Candidate Key Profiles
 - Column Length Distribution Profiles
 - Column Null Ratio Profiles
 - Column Statistics Profiles
 - Column Value Distribution Profiles
 - [dbo].[StgInsurance]
 - Candidate Key Profiles
 - Column Length Distribution Profiles
 - Column Null Ratio Profiles
 - Column Value Distribution Profiles
 - [dbo].[StgRental]
 - Candidate Key Profiles
 - Column Length Distribution Profiles
 - Column Null Ratio Profiles
 - Column Statistics Profiles
 - Column Value Distribution Profiles
 - [dbo].[StgStaff]
 - Candidate Key Profiles
 - Column Length Distribution Profiles
 - Column Null Ratio Profiles
 - Column Statistics Profiles
 - Column Value Distribution Profiles
 - [dbo].[StgVehicle]
 - Candidate Key Profiles
 - Column Length Distribution Profiles
 - Column Null Ratio Profiles
 - Column Statistics Profiles

Candidate Key Profiles - [dbo].[StgCustomer]

Key Columns Key Strength

Customer_ID	100.0000 %
-------------	------------

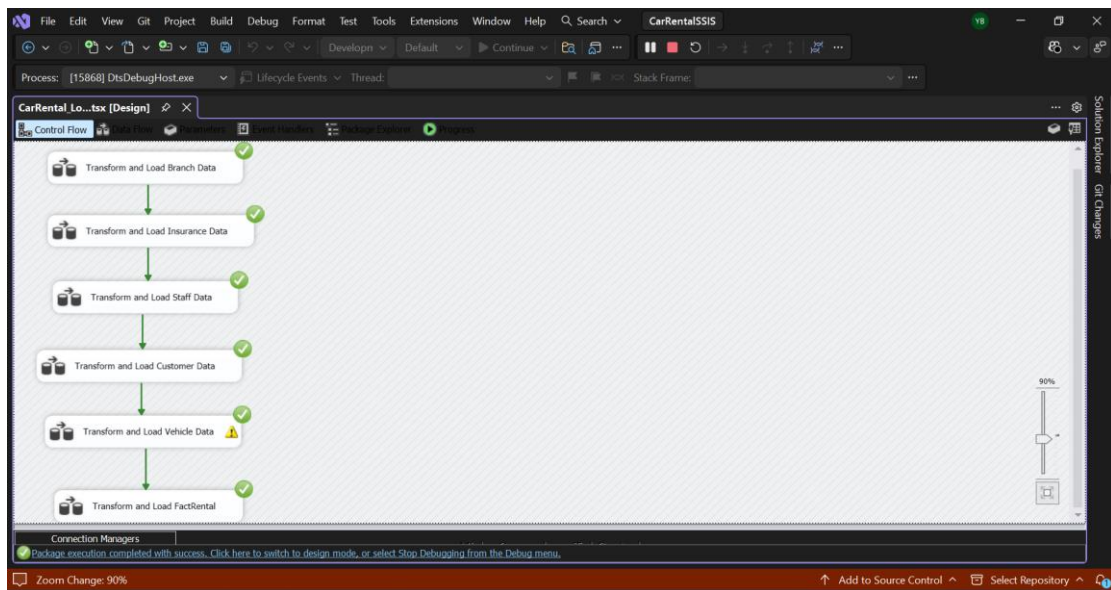
Key Violations

Encrypted Connection 1000 Rows

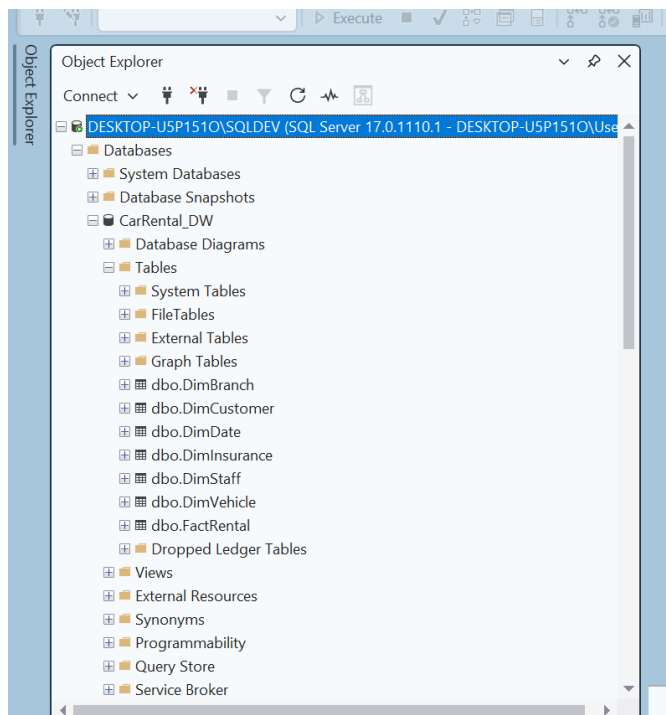
Successfully loaded data profile from ...

Message

Data Transformation

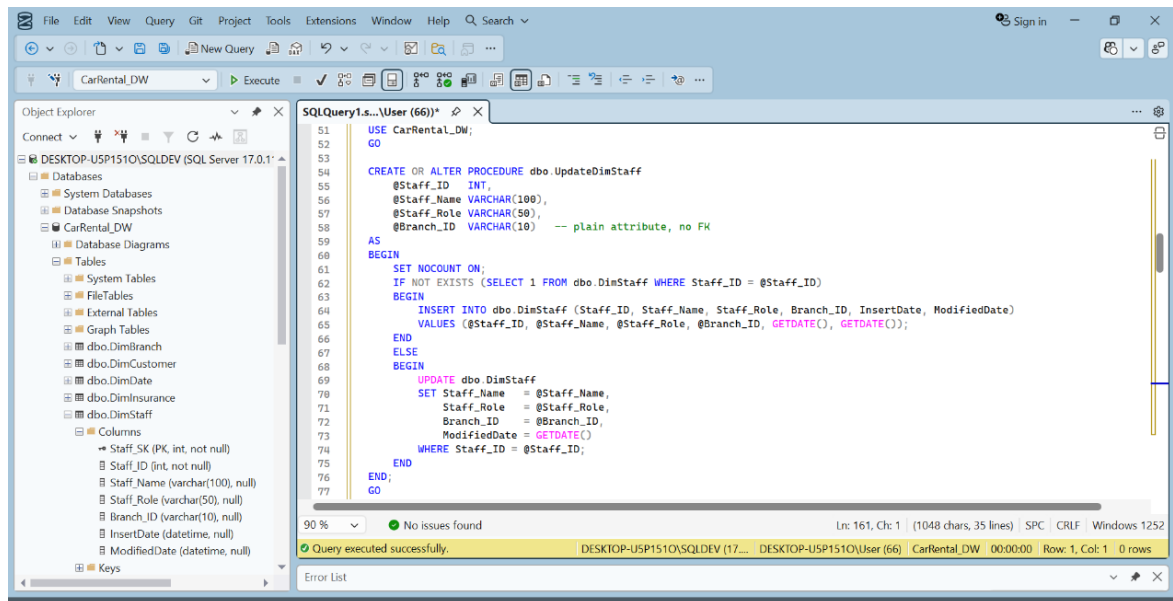


Finally extracted all the data from tables which are in the CarRental_Staging into DB named CarRental_DW database as shown in the below using SQL Server Integration Service Software

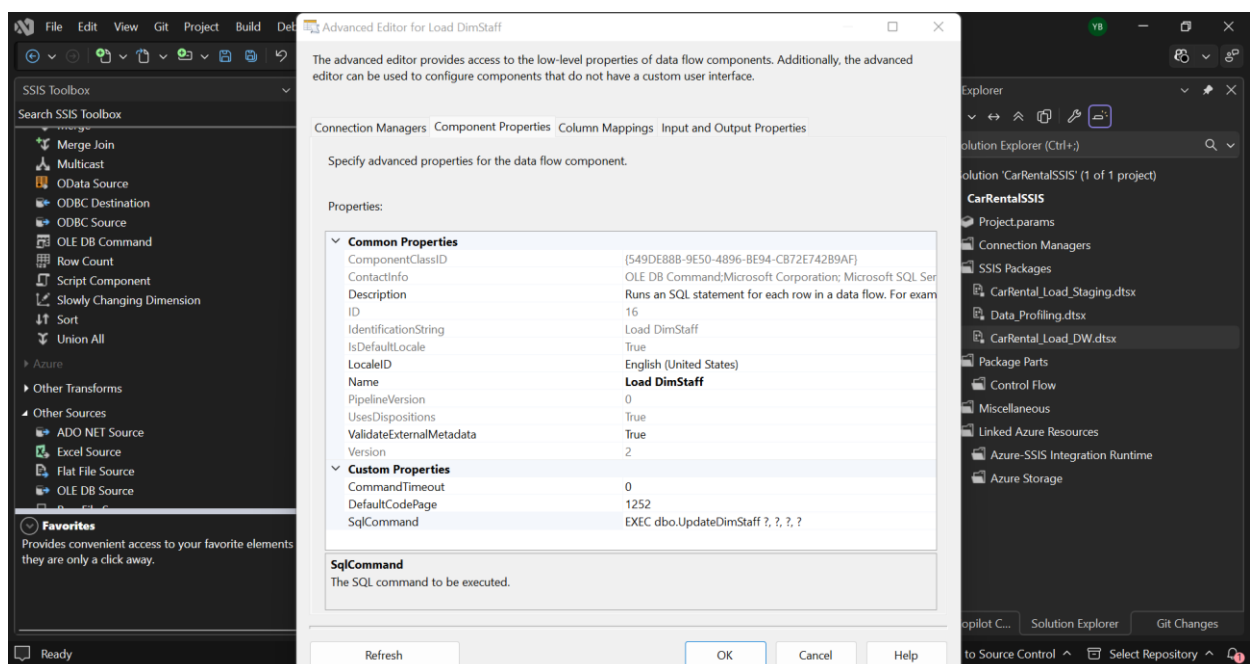


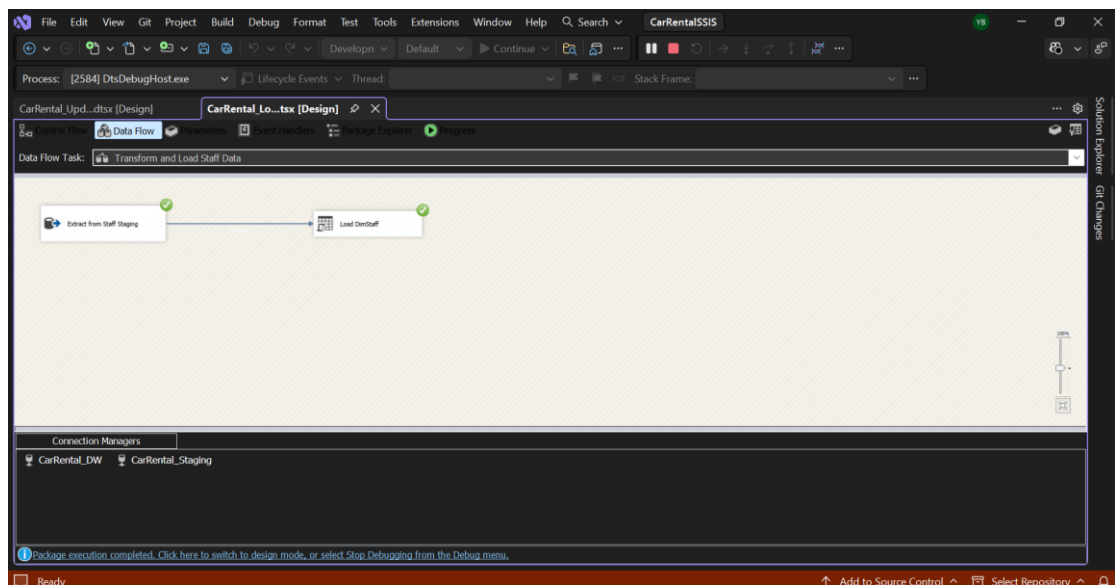
DimStaff

First created a Procedure called UpdateDimStaff and executed in the carRental_DW database



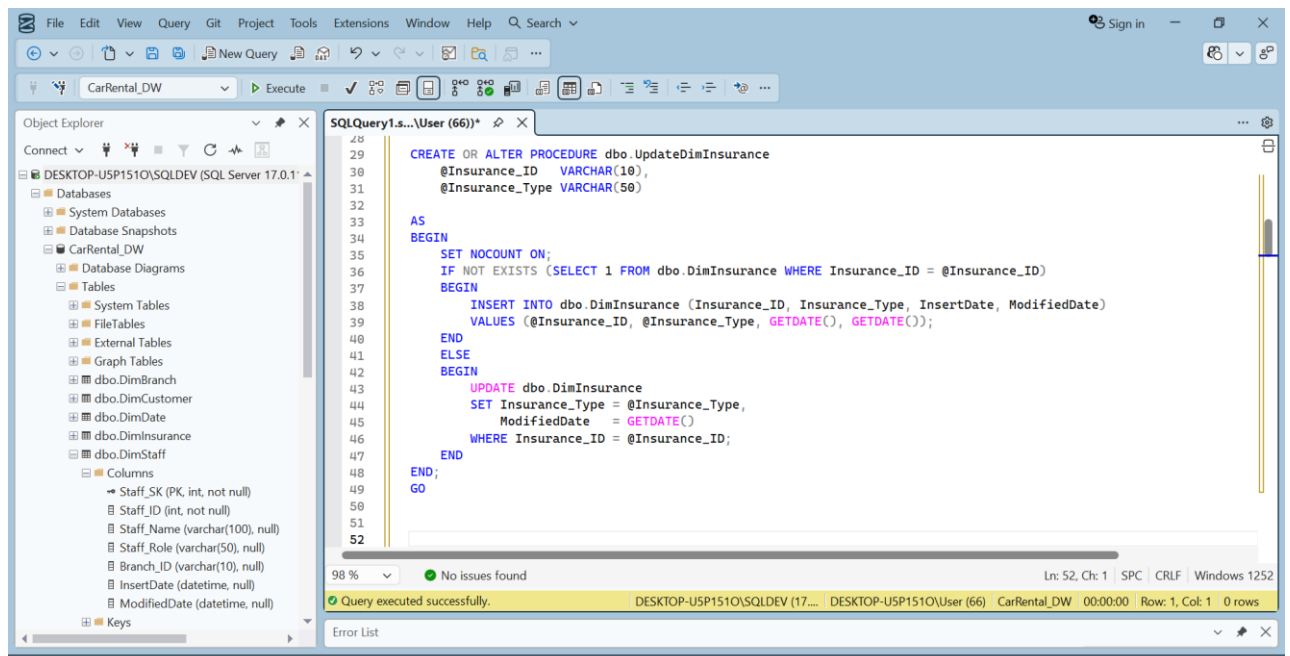
OLE DB Command SSIS tool used to execute, UpdateDimStaff procedure, it used to insert data from Staff details which is in CarRental_Staging to DimStaff table which is in CarRental_DW. Above Stored Procedure ensure no duplicates are entered into the data warehouse table 'DimStaff'. If there is an existing Staff record, it will be updated with the latest record coming from staging table. Else, if it is a new record, it will be inserted into DimStaff



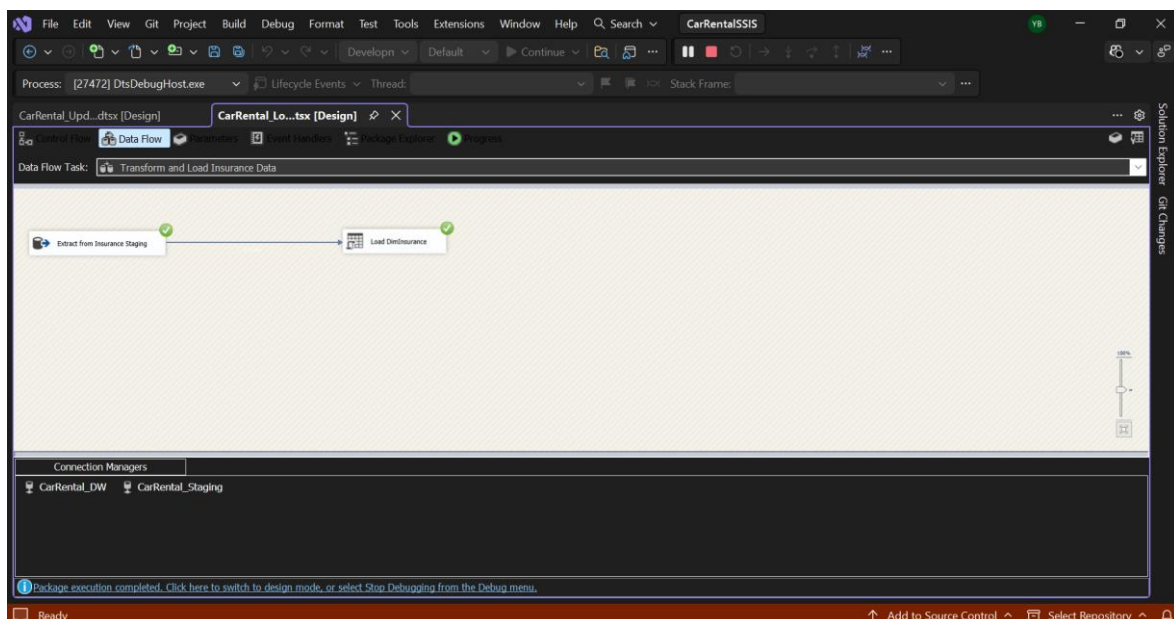
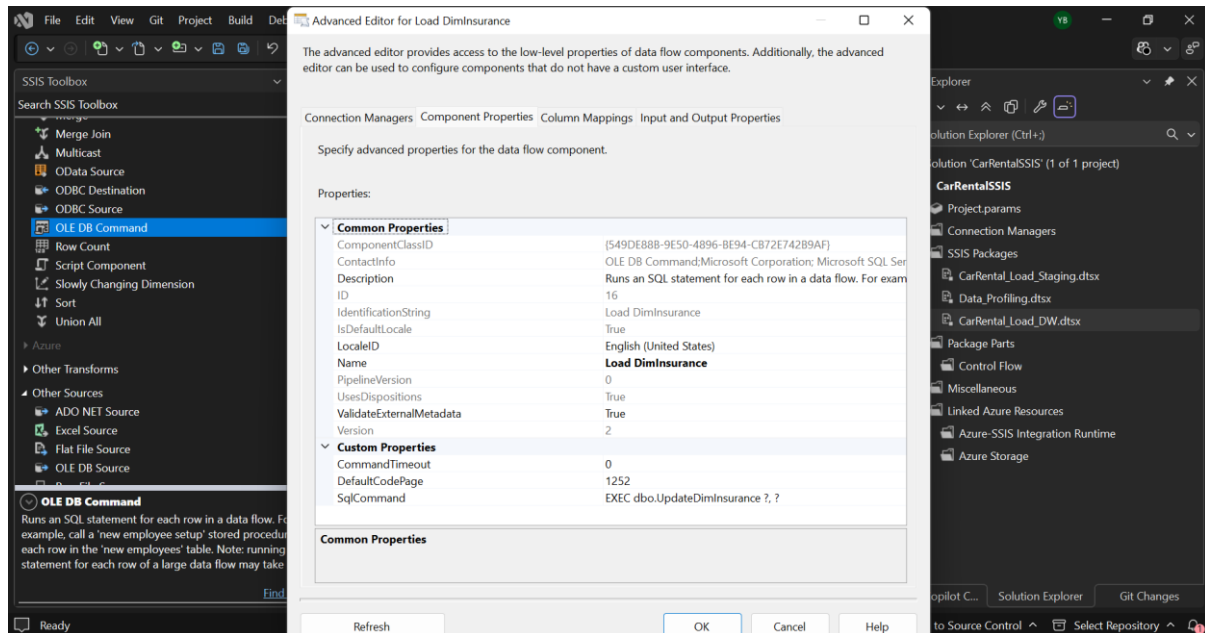


DimInsurance

First created a Procedure called UpdateDimInsurance and executed in the carRental_DW database

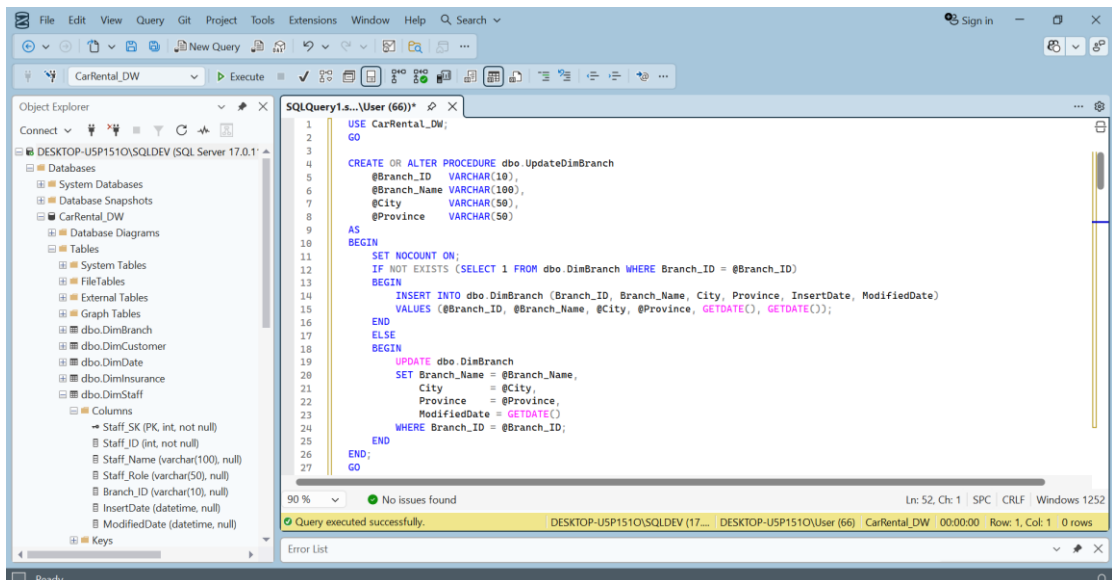


OLE DB Command SSIS tool used to execute, UpdateDimInsurance procedure, it used to insert data from Insurance details which is in CarRental_Staging to DimInsurance table which is in CarRental_DW. Above Stored Procedure ensure no duplicates are entered into the data warehouse table 'DimInsurance'. If there is an existing Insurance record, it will be updated with the latest record coming from staging table. Else, if it is a new record, it will be inserted into DimInsurance

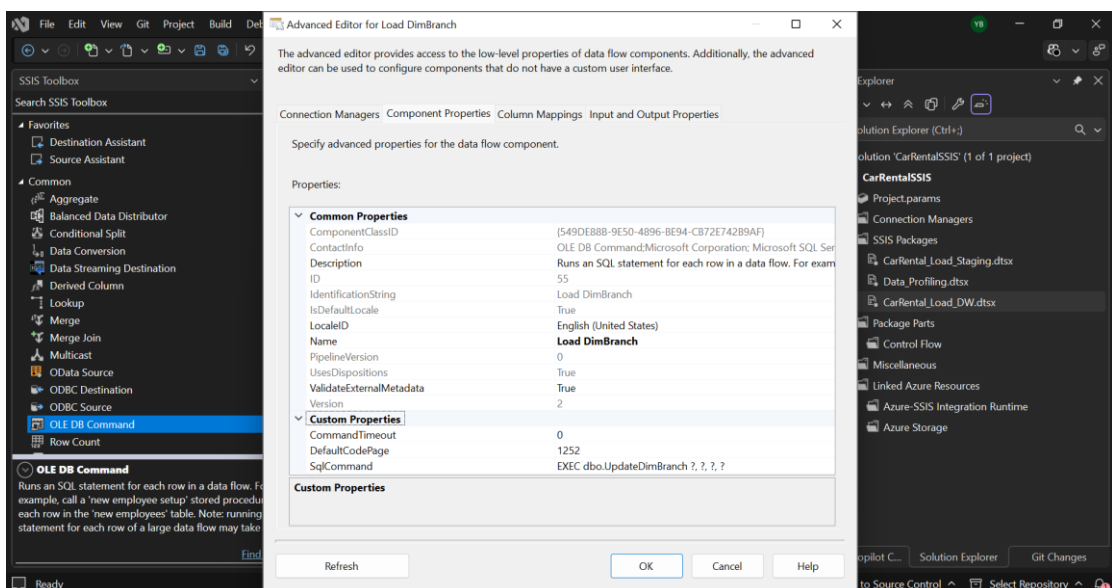


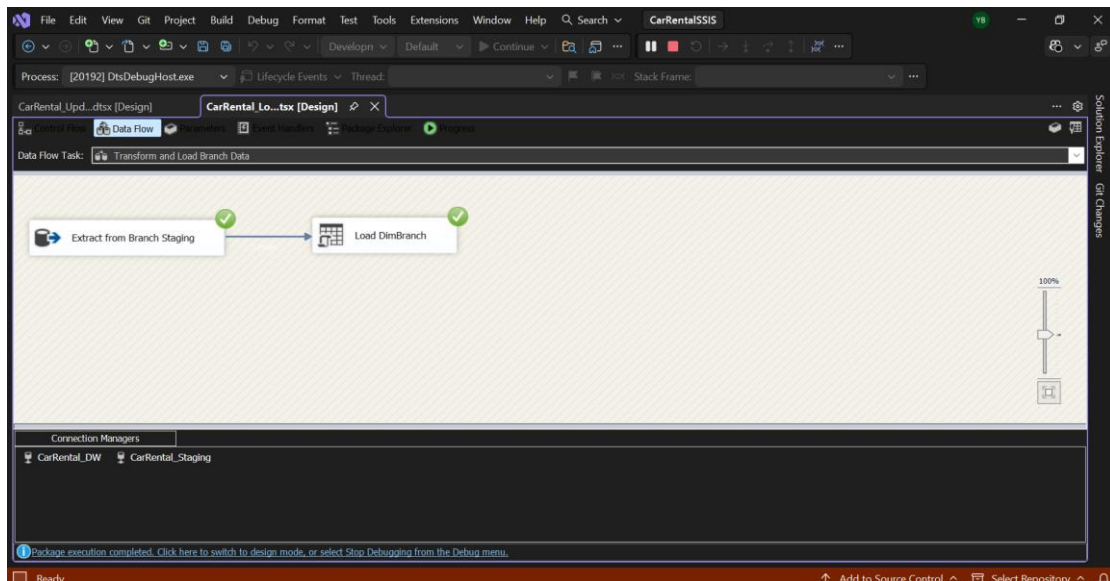
DimBranch

First created a Procedure called UpdateDimBranch and executed in the carRental_DW database



OLE DB Command SSIS tool used to execute, UpdateDimBranch procedure, it used to insert data from Branch details which is in CarRental_Staging to DimBranch table which is in CarRental_DW. Above Stored Procedure ensure no duplicates are entered into the data warehouse table 'DimBranch'. If there is an existing Branch record, it will be updated with the latest record coming from staging table. Else, if it is a new record, it will be inserted into DimBranch

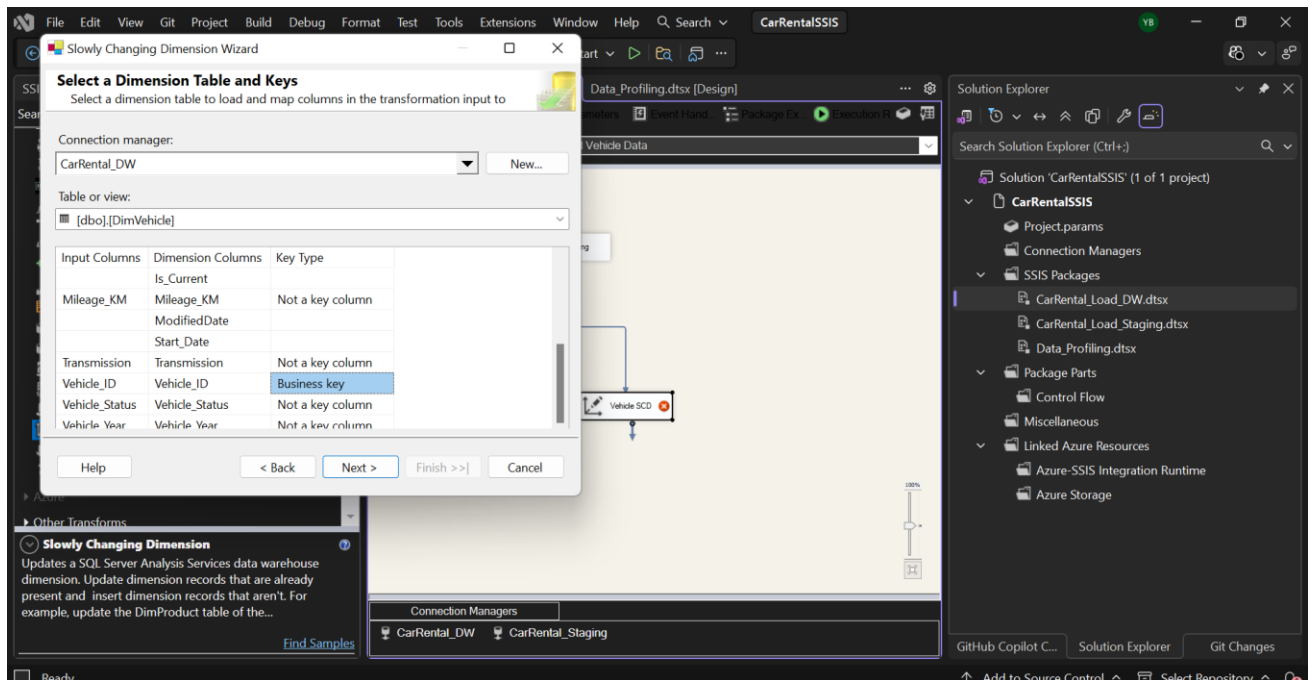


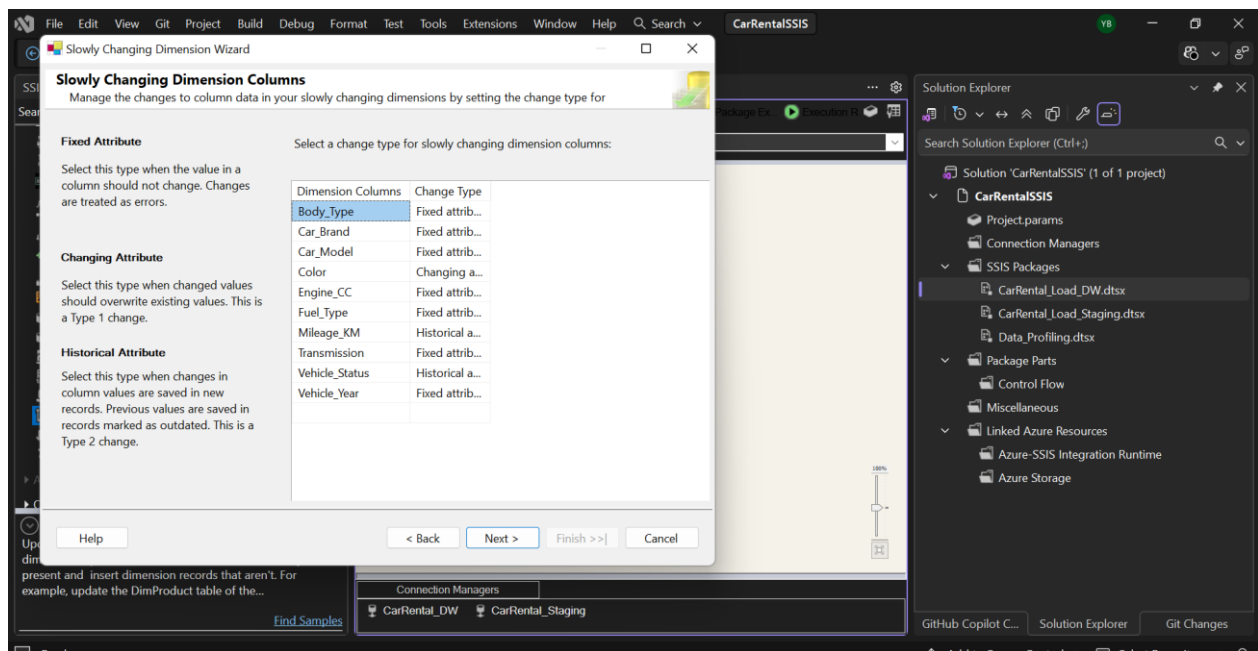


DimVehicle (SCD Type 2)

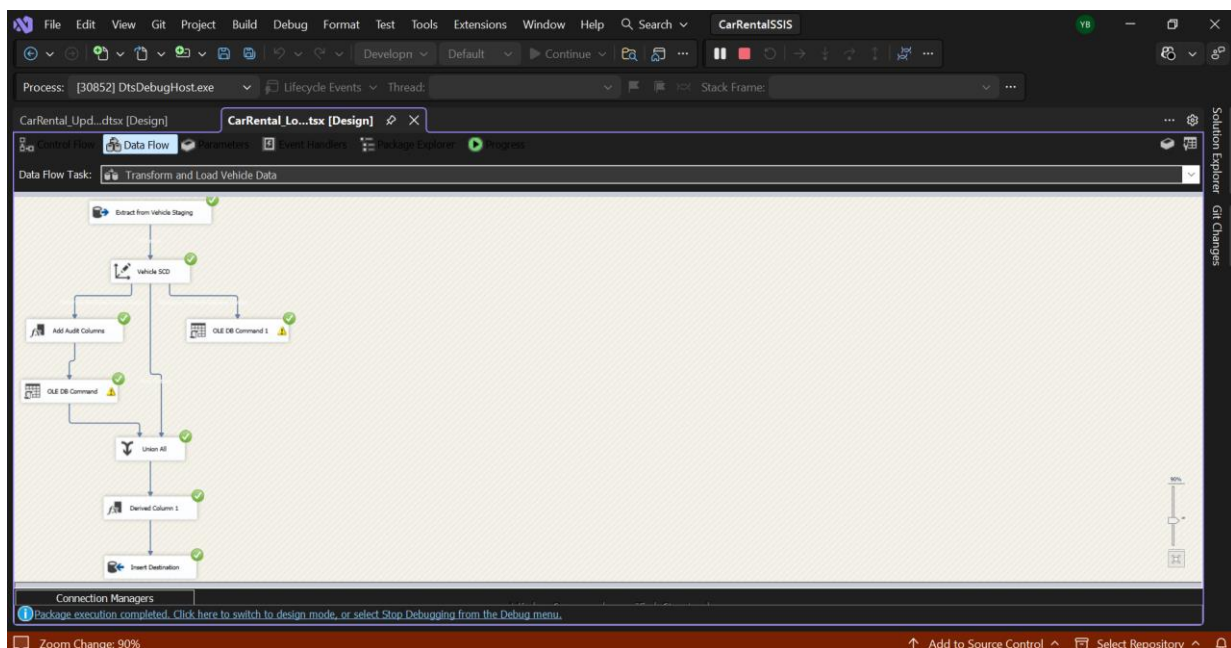
Extract the Vehicle service details from staging and consider DimVehicle as a slowly dimension

In the SCD Configuration Wizard I set the configurations as below



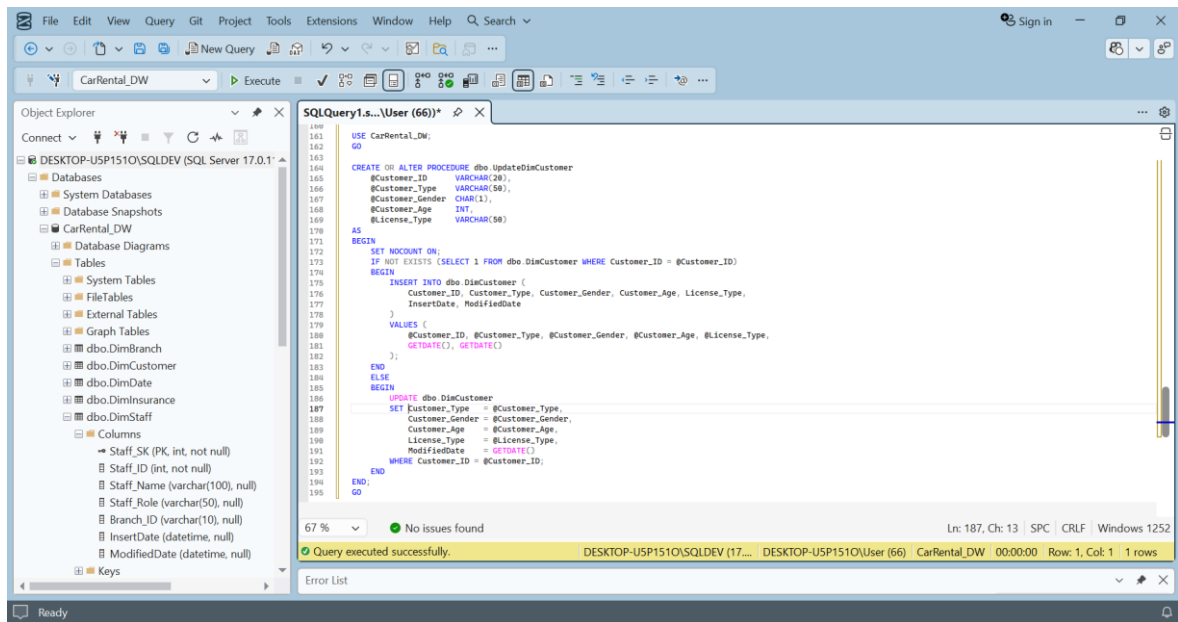


Once All Configurations done properly, it will automatically create the Slowly Changing Dimension as shown below.

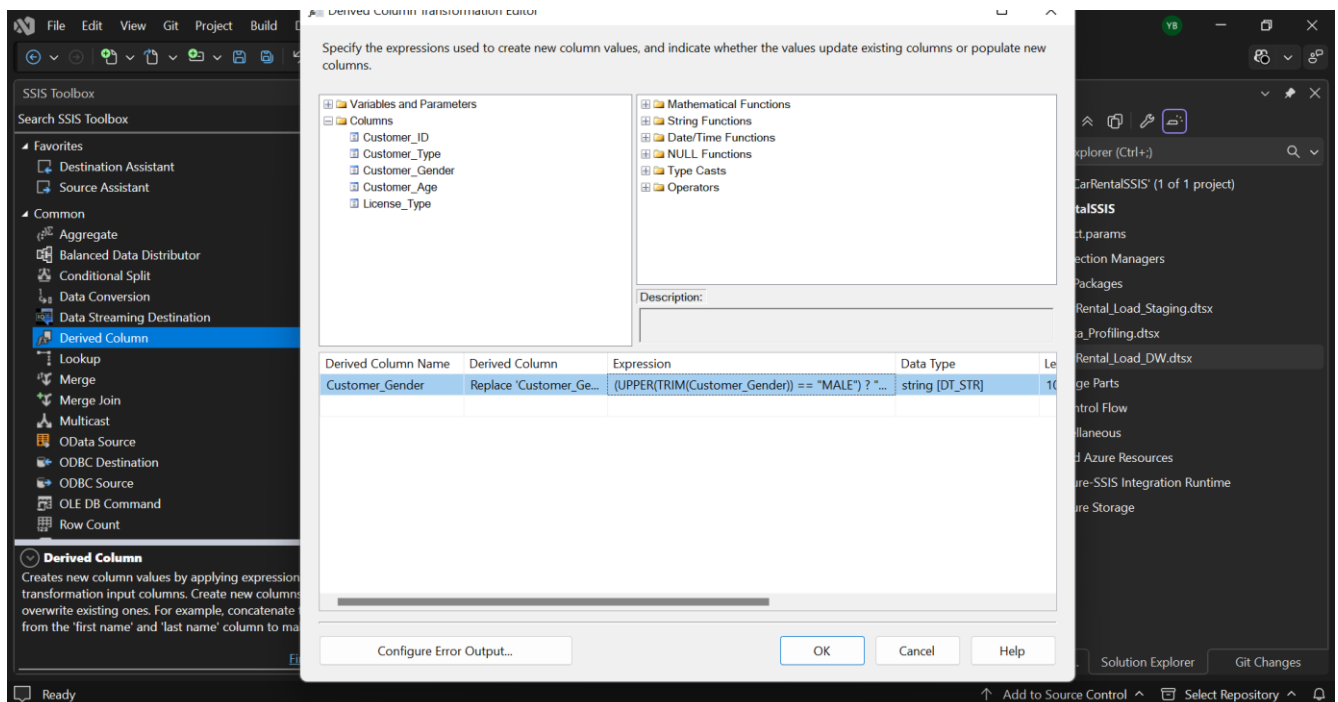


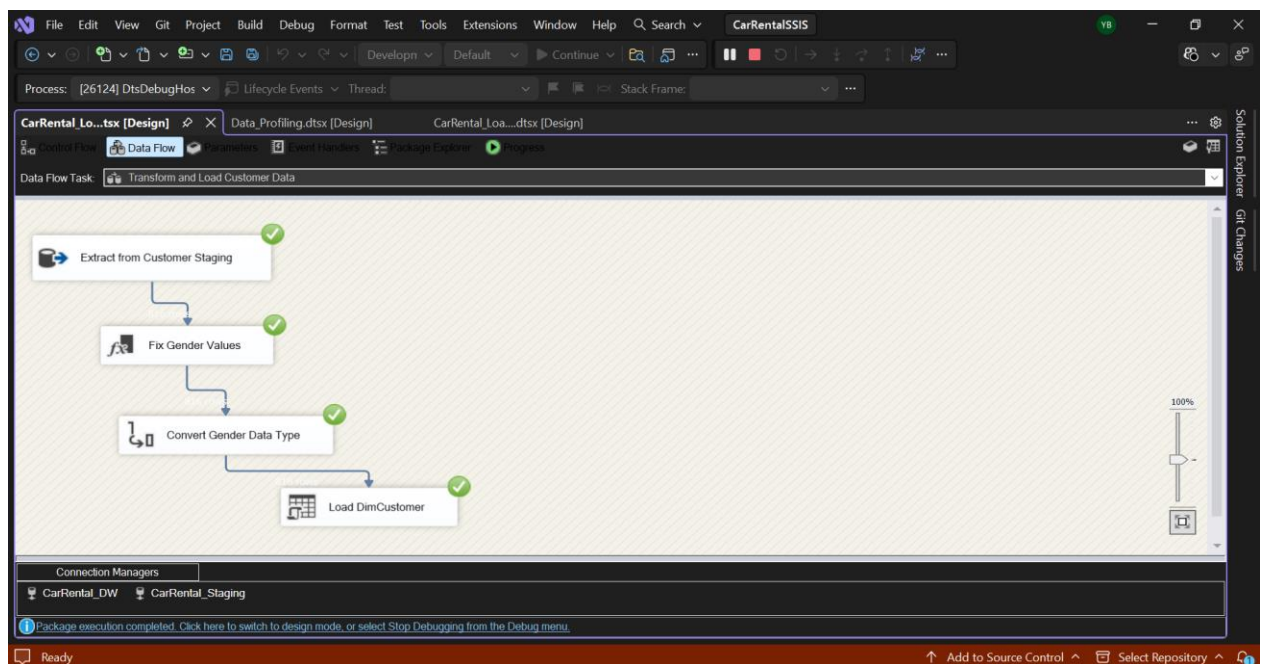
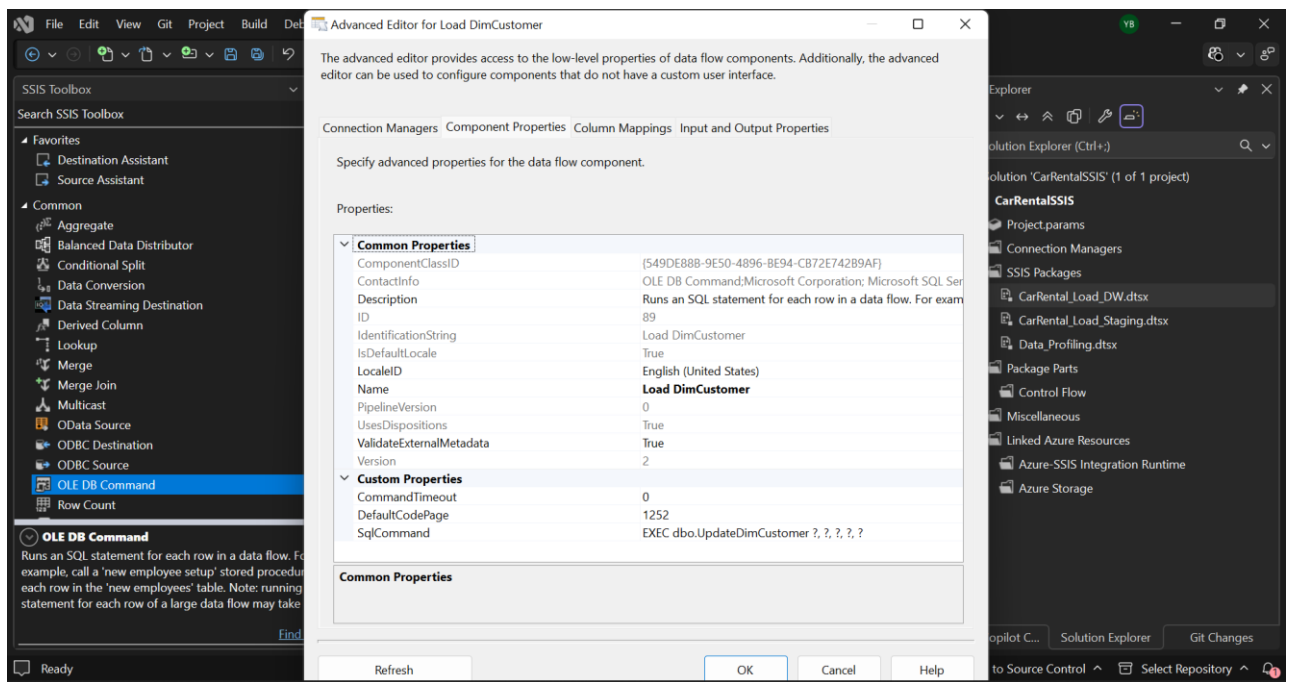
DimCustomer

Created a Procedure called UpdateCustomer and executed in the CarRental_DW database.



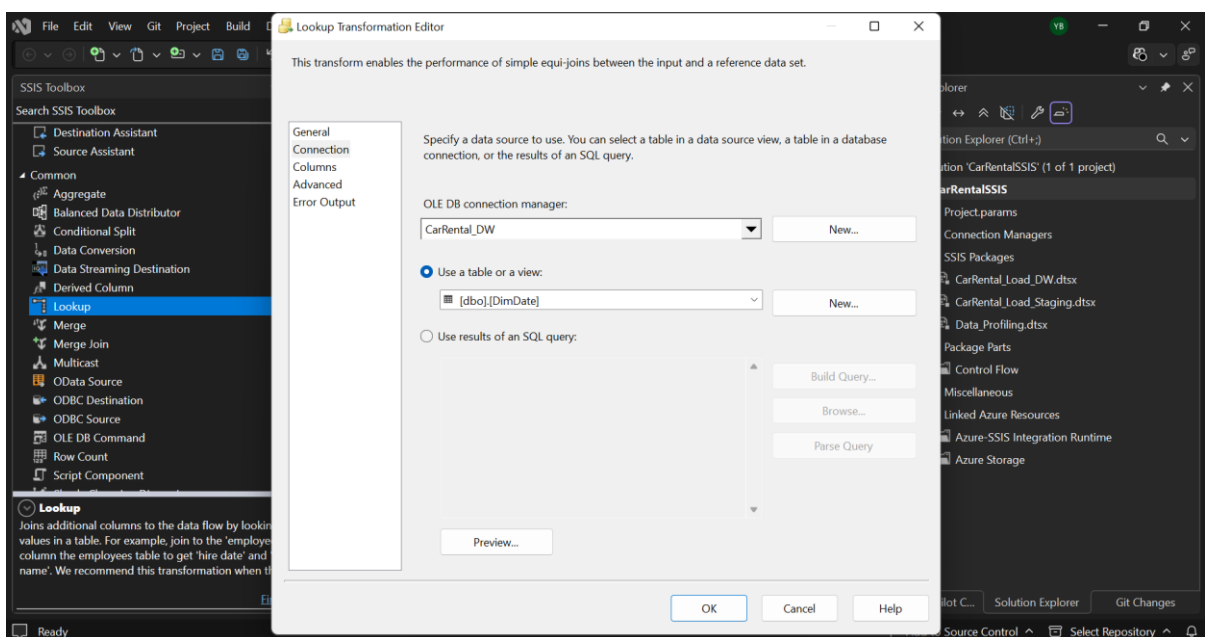
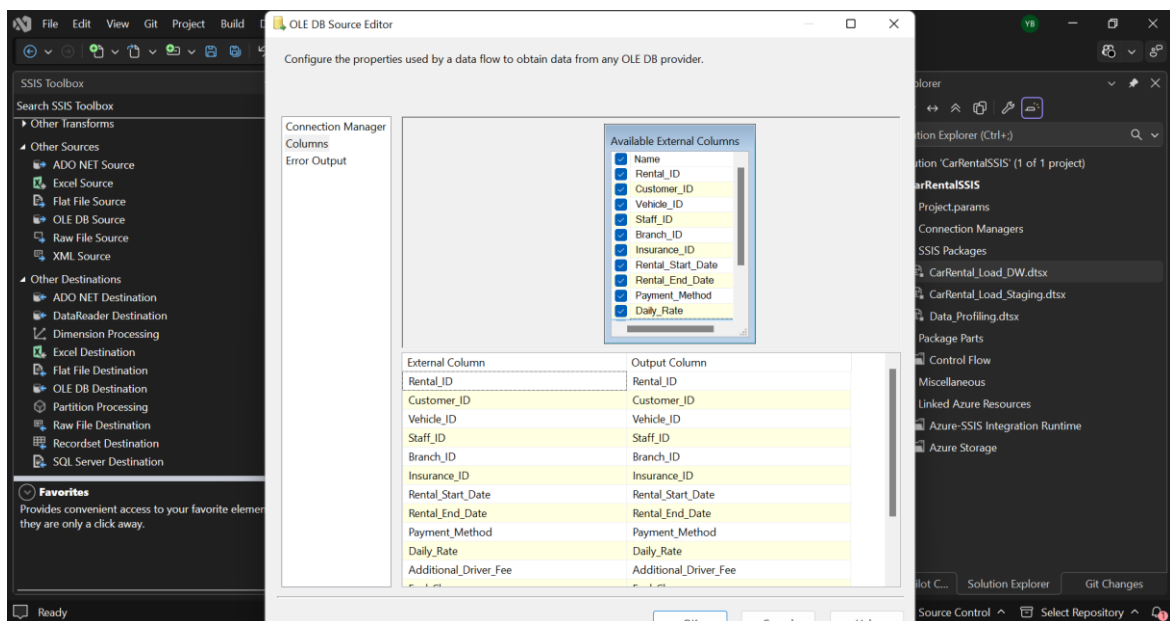
The SSIS package uses a Flat File Source, Data Conversion transformation for correct data types, and OLE DB Destination to load into DimCustomer.

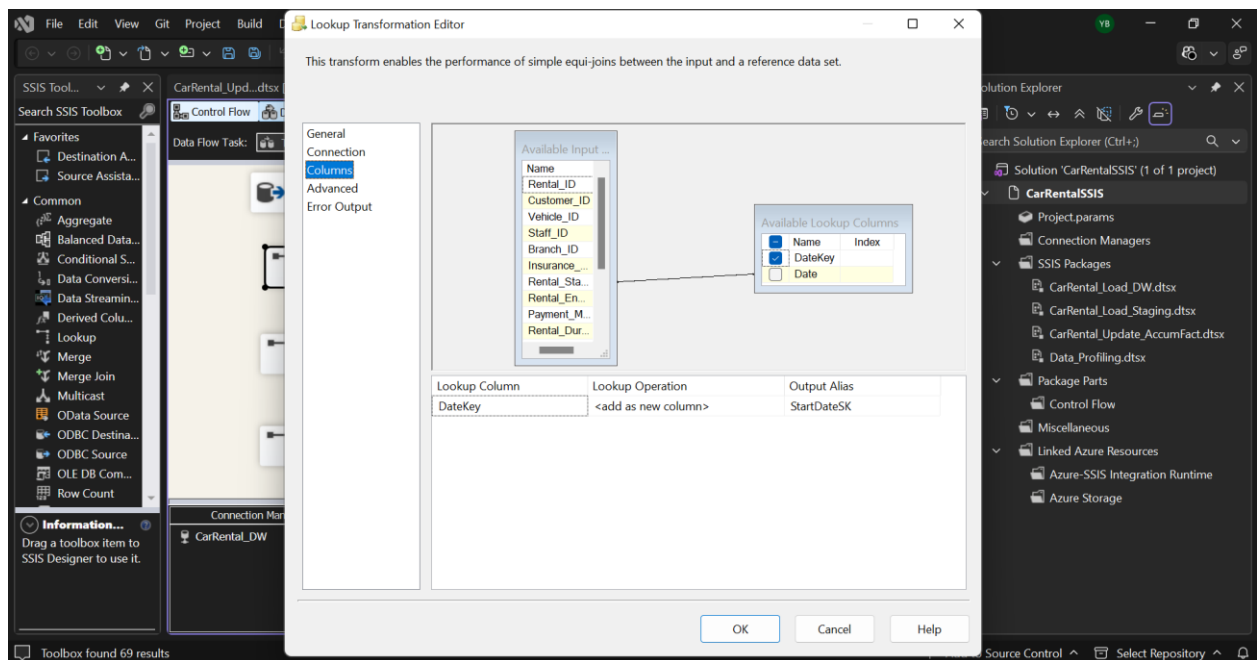




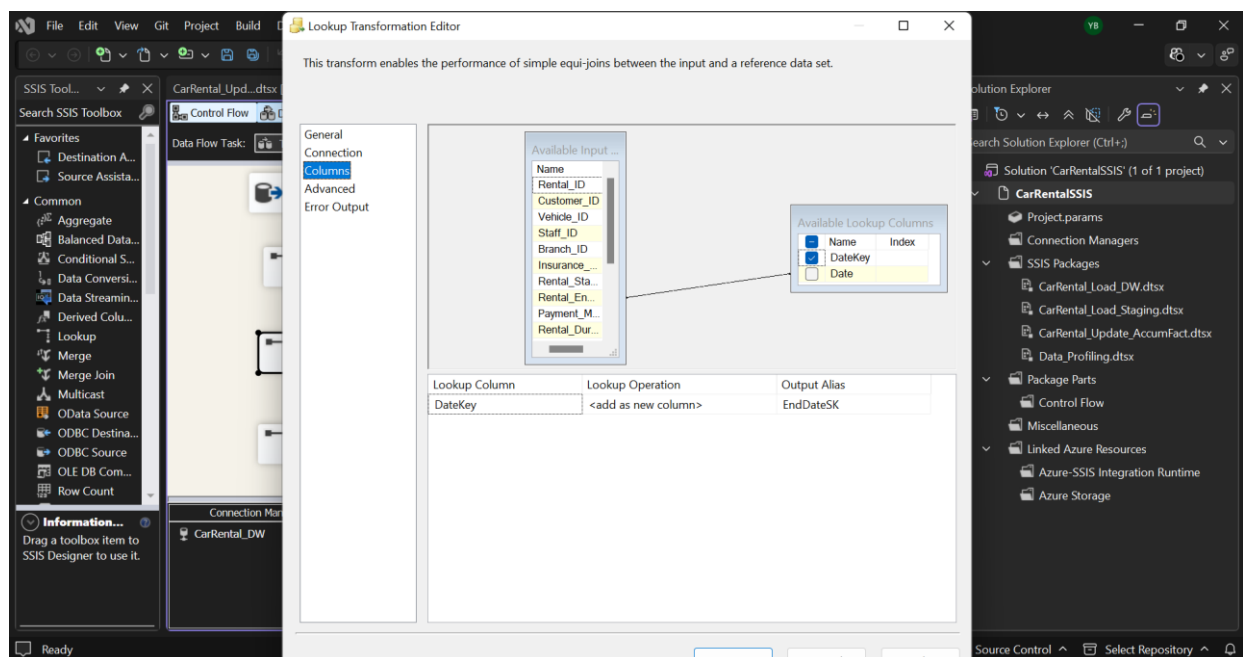
FactRental

The fact table is loaded last after all dimension tables are populated. Flat File Source reads the CSV, multiple **Lookup transformations** retrieve surrogate keys from each dimension table (Customer_SK from DimCustomer, Vehicle_SK from DimVehicle, Branch_SK from DimBranch, Staff_SK from DimStaff, Insurance_SK from DimInsurance, Start_Date_SK and End_Date_SK from DimDate), a **Derived Column transformation** sets accm_txn_create_time to GETDATE(), and finally OLE DB Destination loads all rows into FactRental.

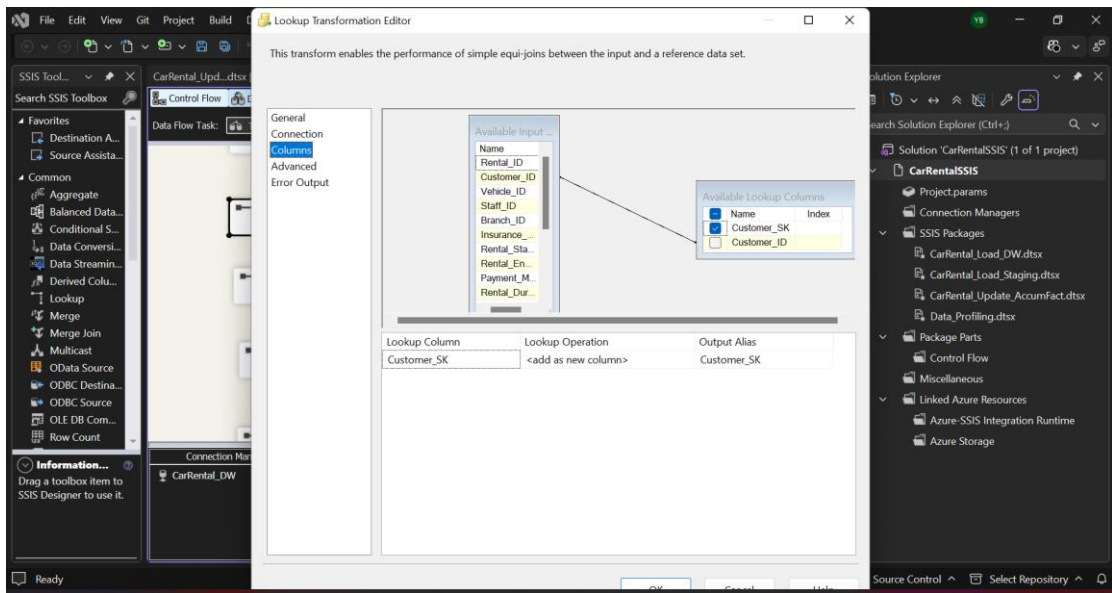




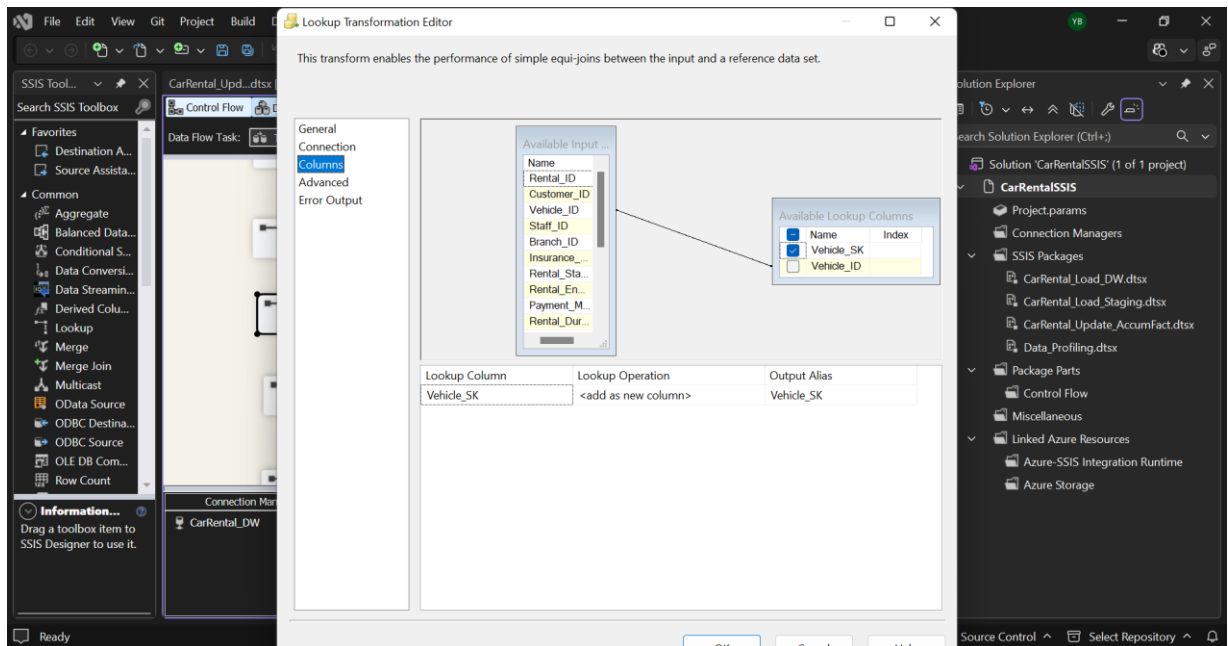
Lookup — DimDate (Start): showing join on Rental_Start_Date, output Start_Date_SK



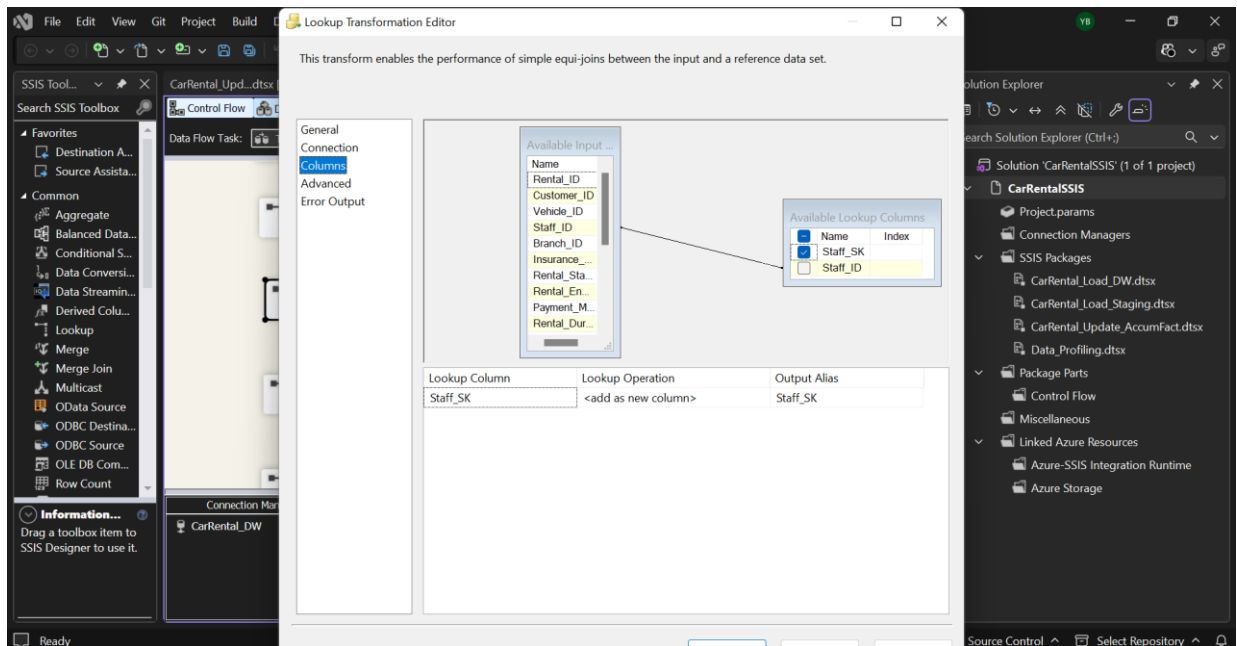
Lookup — DimDate (End): showing join on Rental_End_Date, output End_Date_SK



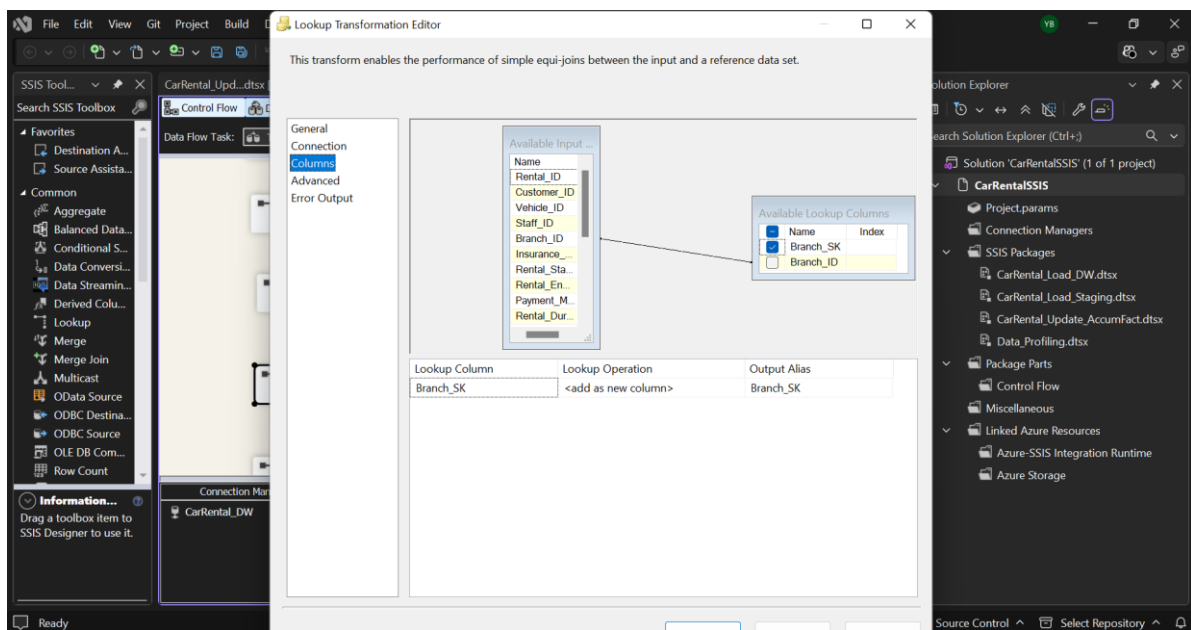
Lookup — DimCustomer: showing join on Customer_ID, output Customer_SK



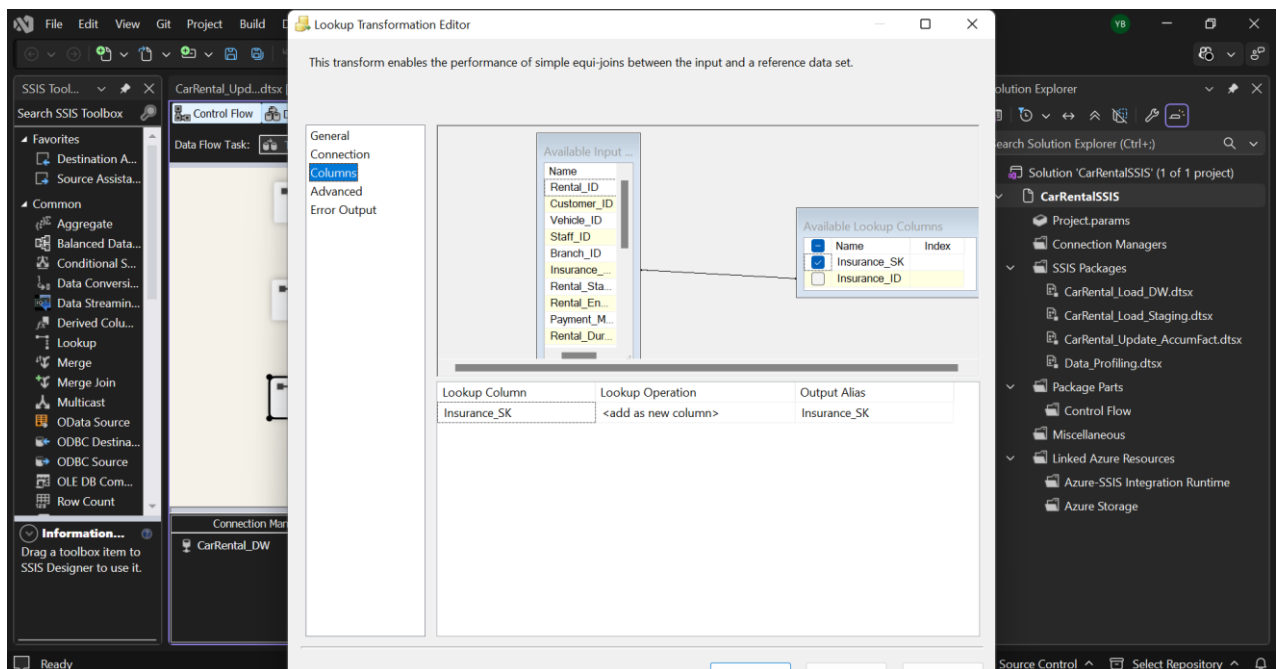
Lookup — DimVehicle: showing join on Vehicle_ID, output Vehicle_SK



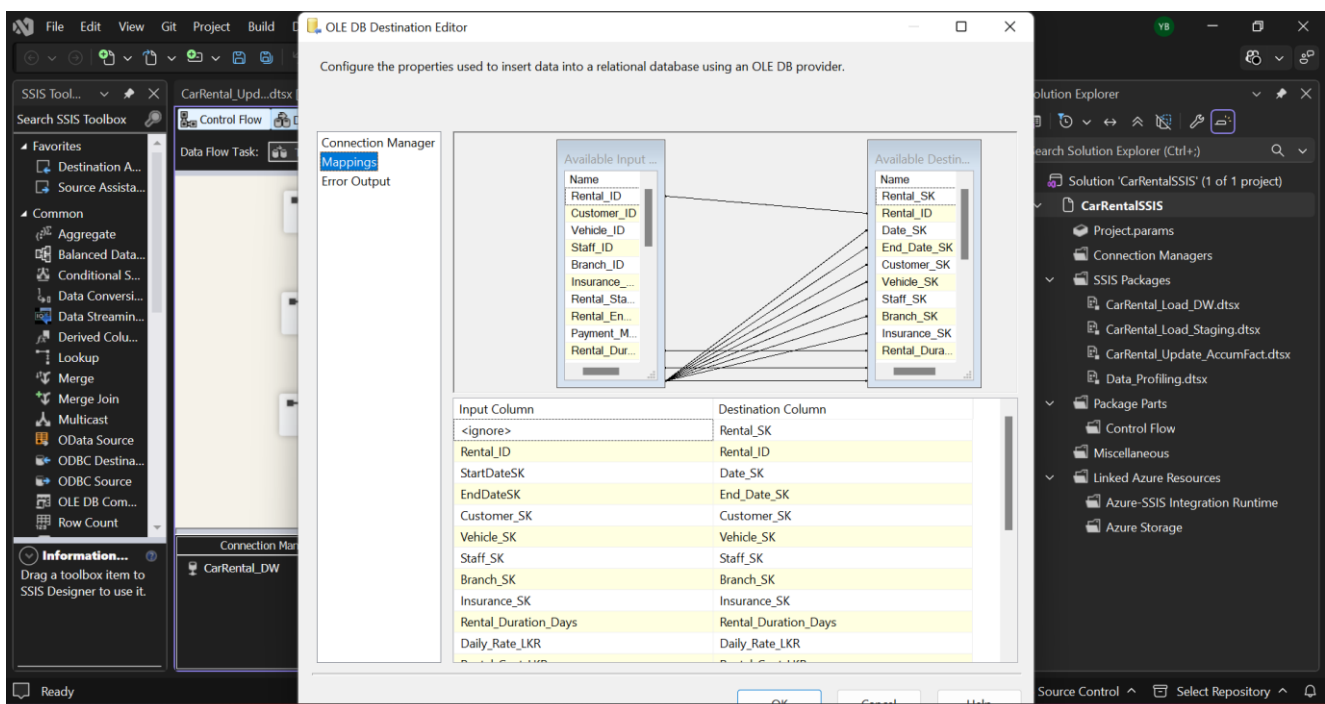
Lookup — DimStaff: showing join on Staff_ID, output Staff_SK

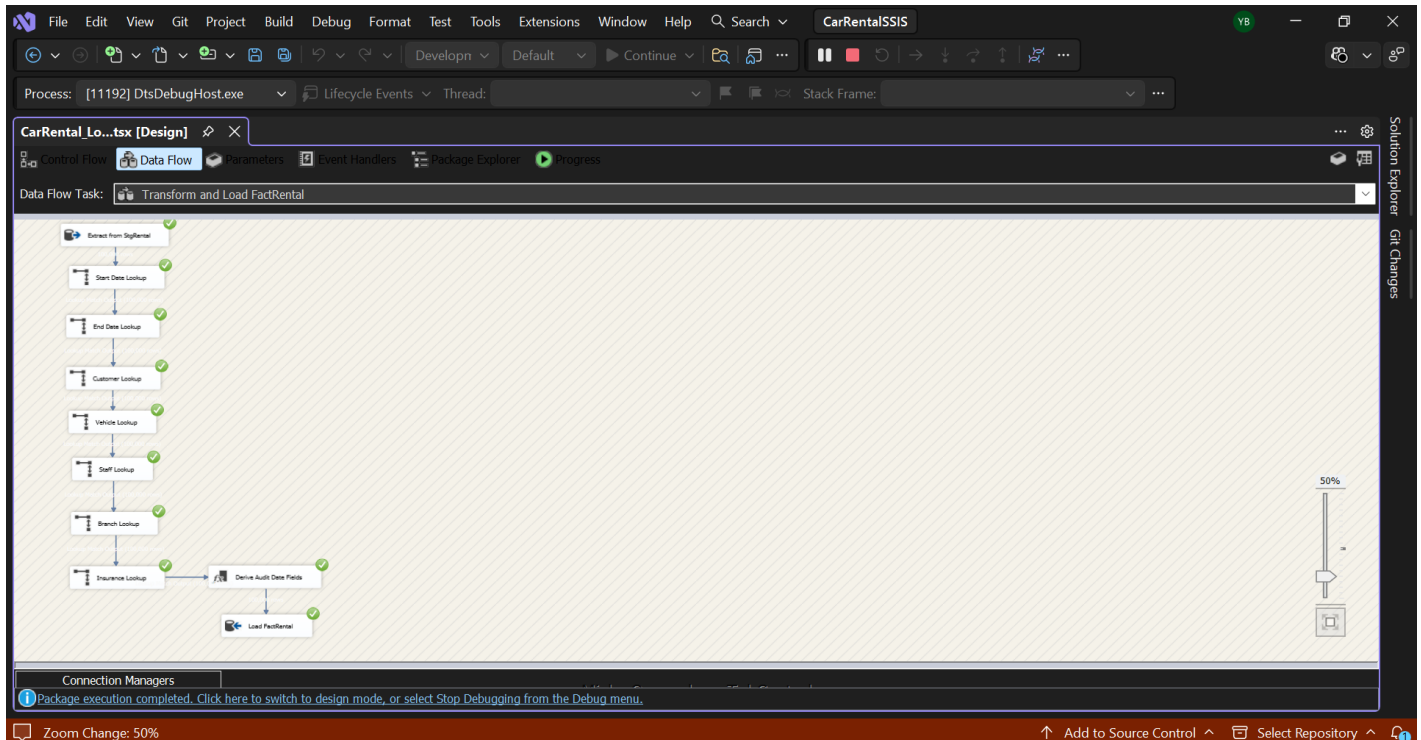


Lookup — DimBranch: showing join on Branch_ID, output Branch_SK



Lookup — DimInsurance: showing join on Insurance_ID, output Insurance_SK





Step 6:ETL development – Accumulating fact tables

I extended my FactRental table by adding three additional columns: accm_txn_create_time, accm_txn_complete_time, and txn_process_time_hours.

During the initial ETL load, accm_txn_create_time was set to the current system date (GETDATE()) for every rental transaction.

A separate CSV file (RentalCompletions.csv) was prepared containing two columns: Rental_ID (natural key) and accm_txn_complete_time. This file simulates the completion timestamp for each rental transaction, which occurs a few days after the rental start.

A dedicated SSIS package named CarRental_Update_AccumFact.dtsx was created. It reads the completion timestamps from the CSV file, looks up the corresponding rental records in FactRental, and updates accm_txn_complete_time. The package also calculates txn_process_time_hours as the difference in hours between accm_txn_complete_time and accm_txn_create_time using the DATEDIFF function.

